

BIOLOGY GRADE 11: REPRODUCTION IN ANIMALS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 3.0: Anatomy and Physiology of Animals
Sub-Strand	Sub-Strand 3.1: Reproduction in Animals
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– External fertilisation in animals – Types of internal fertilisation (viviparous, oviparous and ovoviviparous animals) – Structure of the male and female reproductive systems – Menstrual cycle – Fertilisation, pregnancy and lactation (include the structure of ovum and sperm cell, and role of hormones) – Role of biotechnology in human reproduction (surrogacy, sperm bank, in-vitro fertilisation)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe external fertilisation in animals, b) describe types of internal fertilisation in animals, c) relate the structure of the male and female reproductive systems to their functions, d) describe the process of fertilisation, pregnancy, and birth in humans, e) describe the role of hormones in human reproduction, f) appreciate the role of biotechnology in addressing human reproductive health.
Core Competencies	– Critical thinking and Problem Solving: The learner gives objective ideas to solve reproductive health problems while debating on the role of biotechnology in human reproduction (Lesson 6). – Learning to Learn: The learner seeks information from the resource person while learning about the stages of pregnancy and the process of childbirth from a resource person (Lesson 5). – Communication and Collaboration: Learners work in groups to create classification charts and discuss observations during the dissection and model work (Lessons 1 and 2).
Core Values	– Social Justice: The learner shares responsibilities fairly while making classification charts to group animals based on types of internal fertilisation (Lesson 1). – Peace: The learner resolves differences amicably while debating on the role of biotechnology in human reproduction (Lesson 6).
Science & Engineering Practices	– Asking questions: Learners generate questions about how frogs, hens and cows reproduce and add them to the Driving Question Board. – Developing and using models: Learners build and revise a cumulative model linking fertilisation strategy, anatomy, hormones

	and biotechnology. – Planning and carrying out investigations: Learners observe pond or aquarium life and dissect a small mammal under ethical and biohazard guidelines. – Analysing and interpreting data: Learners interpret hormone graphs of FSH, LH, oestrogen and progesterone across the menstrual cycle. – Obtaining, evaluating and communicating information: Learners use media clips, a nurse or midwife and case studies to evaluate evidence and debate IVF, surrogacy and sperm banking. – Engaging in argument from evidence: Learners debate biotechnology and joint reproductive health choices using scientific and ethical evidence.
Pertinent & Contemporary Issues (PCIs)	– Menstrual Health Disorders Management: The learner is sensitised on human menstrual health while illustrating the role of hormones in the human sexual cycle (Lesson 4). – Human Sexuality: The learner understands common myths and misconceptions around human sexuality while illustrating the role of hormones in the human sexual cycle (Lesson 4).
Career Connections	– Obstetrics and gynaecology: Uses knowledge of reproductive anatomy, hormones and pregnancy care. – Midwifery and nursing: Supports mothers through pregnancy, childbirth and postnatal health. – Embryology: Applies knowledge of fertilisation and early development, as in IVF laboratories. – Biotechnology research: Develops and improves assisted reproductive technologies. – Reproductive endocrinology: Diagnoses and treats hormonal imbalances affecting fertility. – Veterinary medicine: Applies animal reproduction knowledge to livestock such as cows and poultry in Kenyan farms.
Focus for Lessons	This unit explores how animals ensure successful fertilisation and development of offspring, from frogs in ponds and hens on farms to cows and humans, and how science helps when human reproduction fails. Learners predict, observe, explain, track questions on the Driving Question Board and revise a cumulative model each lesson. The approach respects sensitive topics, limited internet access (pre-downloaded clips), shared devices and strict dissection ethics.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? KICD KEY INQUIRY QUESTIONS: 1. How do animals reproduce?
Anchoring Phenomenon	Frogs in local ponds (for example near Kisumu or along Lake Victoria wetlands) release thousands of jelly-covered eggs directly into water, while farm hens lay hard-shelled eggs on land, and cows give birth to live calves after months of internal development. Meanwhile, human couples facing reproductive challenges can use technologies like IVF or surrogacy to have children, as often heard on Kenyan news and social media. It is puzzling that such different strategies all achieve the same goal, why frogs need thousands of eggs while cows produce one calf, and how science can step in when natural reproduction fails (many assume IVF babies grow wholly in a test tube).
Storyline Thread	Lesson 1: External fertilisation requires aquatic environments, whereas internal fertilisation (oviparous, viviparous, ovoviviparous) protects offspring on land. Learners observe pond or aquarium life, search sources on internal fertilisation types and make classification charts. Leads to: What specialised anatomical structures enable mammals to carry out internal fertilisation? Lesson 2: Male and female mammalian reproductive structures are adapted for gamete production and supporting embryo development. Learners dissect a small mammal, draw and label, and use models and charts to discuss adaptations. Leads to: How do human sperm and egg cells meet and interact at the cellular level?

	Lesson 3: Sperm and ovum structures are specialised for movement, protection and fusion during fertilisation. Learners draw and label gametes and present adaptations to peers. Leads to: What controls the monthly release of these gametes and prepares the body for pregnancy? Lesson 4: Pituitary and ovarian hormones interact through feedback mechanisms to regulate the menstrual cycle and gamete release. Learners trace FSH, LH, oestrogen and progesterone on charts and discuss imbalances, myths and menstrual health. Leads to: How does the hormonal balance shift if fertilisation occurs and leads to pregnancy? Lesson 5: Gestation, placental nutrition and delivery are driven by hormonal shifts and maternal-fetal structural interactions. Learners watch animations of fertilisation, implantation, placenta formation and birth, and consult a nurse or midwife. Leads to: What happens when natural fertilisation or pregnancy cannot occur? Lesson 6: Reproductive biotechnologies (IVF, surrogacy, sperm banking) provide medical solutions to structural or hormonal infertility. Learners debate biotechnology using case studies and note ethical and scientific arguments. Leads to: How do joint decisions by both parents affect family health and overall reproductive outcomes? Lesson 7: Reproductive health requires shared decision-making, hormonal health awareness and informed uptake of medical technologies. Learners debate joint reproductive choices, review the Driving Question Board and compile final synthesis models as a unit reflection.
Supporting Phenomena	– Tilapia in Lake Victoria and aquarium fish release eggs and sperm into water, with no mating contact. – Hens on a Kenyan farm lay shelled eggs that need warmth (incubation) to hatch, while cows and goats give birth to live young. – Many girls and women experience a monthly cycle with variations in timing and discomfort, linked to hormones. – A midwife at a local health centre monitors pregnant mothers, showing how the placenta nourishes the foetus. – News stories of couples using IVF, surrogacy or sperm banks raise both hope and ethical debate.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (80 minutes (double lesson)): Frogs, Hens and Cows: Many Ways to One Goal – How Do Animals Reproduce?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon (frogs, hens and cows reproduce in very different ways while IVF and surrogacy help when human reproduction fails) and to help learners figure out that external fertilisation requires aquatic environments, whereas internal fertilisation (oviparous, viviparous, ovoviviparous) protects offspring on land.
Knowledge	a) Describe external fertilisation in animals (amphibians and fish). b) Describe types of internal fertilisation in animals (oviparous, viviparous, ovoviviparous). Learners explain why aquatic animals release many gametes while land animals produce few, protected offspring.
Skills	Asking questions; developing and using models (initial concept map); obtaining, evaluating and communicating information (searching sources, classification charts); analysing and interpreting data (observation notes); constructing explanations using evidence.
Attitudes	Core competencies: Critical thinking and Problem Solving (giving objective ideas on reproductive questions); Learning to Learn (seeking information from varied sources). Value: Social Justice - the learner shares responsibilities fairly while making classification charts to group animals by type of internal fertilisation. PCI: Human Sexuality - understands common myths and misconceptions (e.g. that IVF babies grow wholly in a test tube), discussed respectfully and scientifically.
Key Inquiry Question	How do animals reproduce?
Purpose in Storyline	This lesson opens the unit by surfacing learners' initial ideas and questions through the Driving Question Board. The evidence gathered (external fertilisation needs water; internal fertilisation protects offspring on land) raises the next question: What specialised anatomical structures enable mammals to carry out internal fertilisation? This leads into Lesson 2 on male and female reproductive systems.
Safety Notes	Pond/wetland observation: observe from the edge, wash hands with soap after handling water or aquarium items, do not enter water, avoid touching frogspawn with bare hands, return any organisms to the habitat. No animal harm. Use a pre-downloaded clip where internet is unreliable. Sensitive topics (human sexuality, IVF) are handled in neutral, scientific, respectful language that honours diverse cultural and family backgrounds; no learner is asked to share personal experiences. Groups share textbooks and devices fairly.

B. LESSON OVERVIEW

Learners meet the phenomenon: frogs near Kisumu and Lake Victoria wetlands release thousands of jelly-covered eggs into water, hens lay hard-shelled eggs on land, cows give birth to live calves, and Kenyan news reports IVF and surrogacy for couples facing reproductive challenges. In groups, learners first predict and draw an initial concept map comparing fish, birds and cows, and post questions on the DQB. They then gather evidence by observing external fertilisation in a pond or aquarium (or a pre-downloaded clip of tilapia and frog spawning), and searching available sources (textbooks, charts, clips) on oviparous, viviparous and ovoviviparous animals to build classification charts. In the Explain phase they use Think-Pair-Share and evidence to link environment to fertilisation strategy and number of offspring. The DQB is revisited and the model is revised, ending with a new question about the anatomical structures mammals need for internal fertilisation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction animals external fertilisation	Learners view the phenomenon: photos or a short clip of frogspawn in a Kisumu wetland pond, a hen on eggs, a cow with her calf, and a Kenyan news headline on IVF/surrogacy. Individually (3 min) then in groups of 4 (7 min), they draw an initial concept map comparing how fish, birds and cows reproduce (where fertilisation happens, eggs or live young, how many offspring, who protects them). They write predictions: 'Why do frogs need thousands of eggs but a cow produces one calf?' and 'What do I think happens in a test-tube baby?' They post at least one question per group on sticky notes for the DQB.	Display the images and say: 'Look carefully. What is the same and what is different about how these animals start their young?' WAIT TIME 10-15 seconds before taking any answer. Cold-call 3 learners to describe one observation each, without judging correct or wrong. Say: 'There are no wrong ideas today; I want to see your current thinking.' Circulate with a checklist and note misconceptions (e.g. 'IVF babies grow entirely in a test tube', 'all animals lay eggs'). Prompt groups: 'What makes you think that? What evidence would change your mind?' Introduce the driving question: 'How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails?' Keep language about IVF neutral and factual.	Predict-Observe-Explain (Predict step) combined with initial modelling: drawing a concept map makes prior ideas visible to learners and teacher, creates a baseline to revise later, and generates authentic questions that drive the unit.	Observable indicators: each group has a concept map with at least 3 animals and at least 2 comparison features (e.g. egg vs live birth, number of young); each group has posted at least 1 question. Teacher records on a checklist the misconceptions heard (e.g. 'test-tube baby grows fully in a tube', 'fertilisation always happens inside the body'). Not graded.
Observe Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction animals external fertilisation	Part A (20 min): Groups observe external fertilisation in amphibians and fish in their habitat (school pond, nearby wetland edge, or aquarium with tilapia/guppies) or from a pre-downloaded clip of frogs in amplexus laying spawn and tilapia spawning in Lake Victoria. They record in a table: where eggs are released, where sperm meet eggs, jelly coating, number of eggs, any parental care, and the role of water. Part B (20 min): Groups search shared textbooks, charts and pre-downloaded clips for information on internal fertilisation types and sort examples (hen, snake, lizard, tilapia-like mouthbrooder note, shark/guppy, cow, goat, human, python, chameleon) into oviparous, viviparous and ovoviviparous columns on a	Before going out: set rules ('Observe from the edge, no touching spawn with bare hands, wash hands after'). Ask: 'What do you notice about where the eggs are and what surrounds them?' WAIT TIME 10-15 seconds. Cold-call 4 learners to share one observation. Prompt: 'Why might the jelly coating matter? What would happen to these eggs on dry land?' For Part B, assign roles and say: 'Everyone must have a job, and you swap at the half-way point.' Hand out the criteria: each animal needs evidence from a source. Where devices are few, play the clip once for the whole class, pause at 2 points and ask groups to note data. Circulate and ask: 'Which column does this animal go in, and what is your evidence: where do the eggs	Predict-Observe-Explain (Observe step) and structured information search: first-hand and media evidence about external fertilisation, followed by classification, helps learners notice patterns (water dependence, egg number, protection) rather than memorise lists. Classification charts make criteria for grouping explicit.	Observable indicators: observation table completed with at least 4 recorded features (e.g. eggs released in water, many eggs, jelly layer, no/little parental care); classification chart places at least 8 animals correctly with a stated reason. Peer review: groups swap charts and use a 3-point checklist (correct placement, evidence given, neat/clear) and write one 'star' and one 'wish' comment. Teacher notes who contributes in each group to check fairness of roles.

	classification chart. Roles rotate: reader, recorder, chart maker, presenter, so responsibilities are shared fairly.	develop and how are young born?' Clarify: oviparous = eggs laid, development outside body; viviparous = young develop inside mother and are born live; ovoviviparous = eggs develop and hatch inside mother's body without a placenta-type nourishment from her.		
Explain Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction animals external fertilisation	Learners use Think-Pair-Share to answer: (1) Why do frogs and fish need water for external fertilisation? (2) Why do frogs release thousands of eggs while a cow produces one calf? (3) How does internal fertilisation protect offspring on land? Pairs write a 3-sentence claim-evidence-reasoning explanation using their observation table and charts, e.g. 'In ponds near Kisumu frogs release sperm and eggs into water, where gametes can meet without drying out, but many eggs are eaten or washed away, so thousands are needed.' They compare with their Predict-phase concept map and mark one idea to keep and one to change. Groups briefly present their classification charts (2 min each).	Say: 'Think quietly for one minute, then turn to your partner.' WAIT TIME 10-15 seconds after each question before calling on anyone. Cold-call 5 pairs to share, and ask a different pair: 'Do you agree? What evidence supports or challenges that?' Push reasoning: 'What would happen to frog eggs in a dry field? How does a hen's shell or a cow's womb solve that problem?' Address the IVF question carefully: 'IVF means fertilisation is helped to happen outside the body, but the embryo is placed in the womb to grow; we will study this later.' Correct the misconception that babies grow wholly in a test tube, using neutral language. Record key agreed ideas on the board: external fertilisation = aquatic, many gametes, low survival; internal fertilisation = fewer, protected offspring on land.	Think-Pair-Share with Claim-Evidence-Reasoning: individual thinking time ensures every learner constructs an explanation, pair talk rehearses scientific language, and whole-class sharing exposes alternative ideas. Linking explanations back to the phenomenon of frogs, hens and cows ties evidence to the driving question.	Observable indicators: written explanation contains a claim, at least 2 pieces of evidence from observation/charts, and reasoning linking water or protection to number of offspring. Teacher uses oral probes ('How do you know?') with a 3-level scale: names a pattern only / links pattern to environment / links to survival of offspring. Exit ticket (2 min): 'Name one difference between external and internal fertilisation and give one Kenyan animal example of each.'
Driving Question Board (DQB) Creation  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Groups bring their sticky-note questions to a class chart titled with the driving question and the anchoring phenomenon images (frogspawn, hen, calf, IVF headline). They read each other's questions, cluster them into categories (e.g. 'Why eggs vs live birth?', 'How does IVF work?', 'How are mammals built for internal fertilisation?', 'What is pregnancy like?'), and add a new	Say: 'Place your question where it fits, and tell us why.' WAIT TIME 10-15 seconds before probing. Cold-call 4 groups to read a question and explain its cluster. Ask: 'Which questions can we begin to answer from today's evidence? Which need new evidence?' Mark answered parts with a green dot and leave the rest open. Highlight a question such as 'How does a cow's body carry a	Driving Question Board: a public record of class thinking that validates learner curiosity, shows how today's evidence connects to larger questions, and gives a purpose for each future lesson. Clustering teaches learners to identify patterns in questions.	Observable indicators: every learner contributes at least 1 question; at least 3 question clusters are created; at least 2 questions are tagged as 'partly answered' with a reason. Teacher notes the quality of questions (asks 'why/how' rather than only 'what') and flags questions for later lessons. Exit ticket: 'Write the question you most want answered next and why.'

 READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals external fertilisation	question after the evidence activities. Each learner adds at least one question and votes with a star on the one they most want answered.	calf?' and say: 'This is where we are heading next.' Keep the DQB visible in the classroom. Respect sensitivity: accept all appropriately phrased questions, and offer to handle personal or sensitive questions through an anonymous question box.		
Model Building Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals external fertilisation	Learners revisit their initial concept maps and revise them in a different colour pen. They add: where fertilisation occurs (external vs internal), environment (aquatic vs land), egg or live young (oviparous, viviparous, ovoviviparous), number of offspring, and level of protection. They add an arrow labelled 'Science can assist' with a tentative note on IVF, and a question mark where they lack understanding: 'What body structures allow internal fertilisation in mammals?' Pairs compare revised maps and give one suggestion.	Say: 'Use a new colour to show what you have changed and why.' WAIT TIME 10-15 seconds, then cold-call 3 learners to describe one change and the evidence that caused it. Probe: 'Which of your first ideas stayed the same?' Introduce the next step: 'We know internal fertilisation protects young on land. But what specialised structures do mammals like cows and humans have that make this possible?' Close by pointing to the DQB, and ask for 2 quick volunteers to restate what was learned today. Collect maps and exit tickets.	Concept Mapping with cumulative model revision: comparing the initial and revised maps makes conceptual change visible, helps learners see how evidence altered their thinking, and sets up questions that drive Lesson 2.	Observable indicators: revised map shows at least 3 new or corrected links (e.g. external fertilisation linked to water; viviparous linked to internal development and live birth); at least 1 misconception corrected; a new question about structures is recorded. Peer feedback uses 'two stars and a wish'. Teacher scores maps with a 4-point rubric (labels, links, use of evidence, questions).

D. TEACHER REFLECTION

Which learners still hold the idea that IVF babies grow wholly in a test tube, or that all animals lay eggs? Did every group member take a role in the chart making, and was wait time actually kept at 10-15 seconds? Which DQB questions were sensitive, and how can I handle them respectfully in later lessons? Was pond access or pre-downloaded clip use equitable across groups? What adjustments are needed so shared textbooks and devices do not limit participation?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe about how frogs and fish release eggs and sperm in water, and how hens, snakes, cows and other animals
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	were grouped as oviparous, viviparous or ovoviviparous?
What did I learn?	What did we learn about why external fertilisation needs water and produces many eggs, while internal fertilisation protects fewer offspring on land?
How does this explain the phenomenon?	How does this help explain why frogs release thousands of eggs while a cow produces one calf, and what new question do we have about the structures mammals need for internal fertilisation and how IVF fits in?

LESSON 2 (80 minutes (double lesson)): Built for Reproduction: Dissecting Mammalian Reproductive Structures

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners relate the structure of male and female mammalian reproductive systems to their functions by dissecting a small mammal and using models and charts, so they can explain how internal development in cows (one calf) differs from the frog strategy of thousands of eggs released into the water.
Knowledge	Learners relate the structure of the male and female reproductive systems to their functions (KICD SLO c). They identify the testes, epididymis, sperm duct, urethra, ovaries, oviducts, uterus, cervix and vagina, and explain how each is adapted for gamete production, transport, fertilisation or supporting the embryo.
Skills	Dissecting and observing a small mammal; drawing and labelling biological diagrams; using models and charts to compare structures; predicting, observing and explaining (Developing and Using Models, Planning and Carrying Out Investigations, Constructing Explanations, Engaging in Argument from Evidence).
Attitudes	Respect for animal life and welfare; teamwork and fair sharing of roles (Social Justice); peaceful handling of differing views (Peace); maturity and respect when discussing human sexuality and diverse cultural backgrounds. Core competencies: Critical Thinking and Problem Solving; Learning to Learn.
Key Inquiry Question	How do animals reproduce?
Purpose in Storyline	Lesson 1 showed that external fertilisation (frogs, tilapia from Lake Victoria) needs thousands of gametes because many are lost, while internal fertilisation saves gametes. Lesson 2 shows the internal structures that make this possible in mammals such as cows and humans. It leaves the open question: How do human sperm and egg cells meet and interact at the cellular level? This leads to Lesson 3.
Safety Notes	Use only a specimen obtained ethically and legally (for example a preserved or humanely euthanised laboratory rat/rabbit supplied by the school, or a prepared specimen; never an animal killed for the lesson). Teacher alone handles the specimen preparation. Learners wear gloves, aprons and goggles, and tie back hair. Scalpels and scissors are used only in groups of 4 with one handler and one recorder at a time. Keep food and drink away. Wash hands with soap after. Dispose of tissue by burying or by school protocol. Learners with objections may take the observer, recorder or model role without penalty. Use anatomical models and charts if no specimen is available. Keep language scientific and respectful, and give students the right to pass on any discussion.

B. LESSON OVERVIEW

Learners begin with the phenomenon: a cow gives birth to one calf after months inside the mother, whereas a frog near Kisumu releases thousands of eggs. They predict what internal structures a mammal needs to make this possible, then dissect a small mammal in groups of 4, supported by models and charts, to draw and label the reproductive system. In the Explain phase they use Think-Pair-Share and a concept map/hormonal flowchart to link structure to function, and a short structured debate on which adaptations matter most. They add questions to the Driving Question Board and revise their class model, which leads to the next question about how sperm and egg meet.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction animals mammalian reproductive  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: reproduction animals mammalian reproductive	(10 min) Learners look at a photo or sketch of a Friesian cow with a calf from a Kericho dairy farm beside a frog pond near Kisumu with jelly-covered eggs. Individually, they sketch on a sheet what they think is inside a female and male mammal that lets one embryo develop safely inside the mother, and label at least 3 parts they expect. They mark each prediction with a confidence tick (sure / not sure). They also predict: Do the male and female systems look similar or different, and why? They share predictions in pairs.	Display the images and say: 'Both the frog and the cow achieve the same goal. Take 2 minutes, in silence, to sketch what you think is inside a mammal that makes it possible for a calf to develop for nine months.' Circulate and look for misconceptions (for example that the embryo grows in the stomach, or that urine and sperm use separate tubes). After the sketch say: 'Turn to your partner and compare.' Allow WAIT TIME of 10-15 seconds after each question before calling on anyone. Cold-call 4 pairs to share one predicted structure and one reason. Record all predictions on the board without judging. Remind learners of the lab rules and the respect expected when discussing reproduction: 'We are scientists studying living things, so we use scientific names and treat the specimen with respect.'	Predict-Observe-Explain (Predict step): committing to a prediction in writing exposes prior ideas and makes learners curious to check them against evidence during the dissection. Comparing a confidence level with later evidence helps them see how their thinking changes.	Observable indicators: (1) predictions include at least 3 labelled parts; (2) learners give a reason that connects structure to function, such as 'the uterus is stretchy so the baby can grow'; (3) the teacher tallies on the board how many predict a separate organ for embryo development, and notes misconceptions (such as 'IVF babies grow wholly in a test tube' or 'embryo grows in the stomach') for later revisiting.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction animals mammalian reproductive  READING: No closely matching reading in the ARES library for this activity. Search ARES	(30 min) Part A: In groups of 4 (roles: handler, reader, recorder, safety monitor, rotated at half time), learners dissect a small mammal (such as a rat or rabbit) following a pre-teacher-demonstrated procedure. Male specimen groups locate testes, epididymis, sperm duct, urethra and penis. Female specimen groups locate ovaries, oviducts (fallopian tubes), the Y-shaped uterus, cervix and vagina. Groups swap observations with another group that had the opposite sex, or use anatomical models and charts to see the other system. They measure and note the size, shape, texture and position of each organ. Part B: Each learner individually	Before starting, check consent and role allocation, and say: 'Safety first: gloves, aprons, goggles on. Only the handler touches the scalpel. If you are uncomfortable you may be the reader or recorder.' Demonstrate the first incision on one specimen for 5 minutes, naming each structure with scientific words. During the work, move group to group and ask: 'What do you notice about the thickness of the oviduct wall?' 'Why might the uterus have two horns in a rat or rabbit but one chamber in a human?' 'Where does the sperm duct meet the urethra?' Allow WAIT TIME of 10-15 seconds after each question. Cold-call 3 groups to report one	Predict-Observe-Explain (Observe step): first-hand dissection data, combined with models and charts, gives learners concrete evidence to test predictions. Drawing and labelling forces them to notice details (for example the coiled epididymis, the muscular cervix) that they will later link to function.	Annotated dissection drawing checked with this checklist: (a) at least 6 structures correctly labelled; (b) accurate relative position and shape; (c) a function note for each organ; (d) a title and use of a ruled line for each label. Safety observation checklist: gloves worn, scalpel passed safely, clean-up done. Oral probe: learners accurately name where sperm are made (testes) and where an embryo develops (uterus).

for readings Search terms: reproduction animals mammalian reproductive	draws and labels the reproductive system with a title, magnification note and a one-line function annotation for each organ. They also draw how it compares to the cow (chart) and the frog (Lesson 1 notes).	surprising observation. Ask the safety monitor of each group to confirm clean-up. Hold up the human model and chart and say: 'Compare what you see with the human model. What is the same? What is different?' Ensure all tissue is disposed of correctly and hands are washed.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: reproduction animals mammalian reproductive  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: reproduction animals mammalian reproductive	(20 min) Learners use their drawings to complete a structure-function table: organ, structure observed, function, adaptation (for example: testes outside the body keep sperm cooler; epididymis coiled to store sperm; ovary produces eggs and hormones; oviduct has cilia to move the egg and is the site of fertilisation; uterus has a thick muscular wall and rich lining for embryo implantation; cervix is a muscular seal that protects the embryo). With a partner they do Think-Pair-Share, then build a concept map/hormonal flowchart linking the hypothalamus, pituitary (FSH, LH), ovary/testis, oestrogen, progesterone and testosterone to gamete production and a supported pregnancy. In a short structured debate (two sides, 8 min), learners argue: 'Which adaptation is most important for the survival of offspring: producing many gametes (frog) or protecting a single embryo inside the mother (cow/human)?' using evidence from the dissection and Lesson 1.	Say: 'Think silently for one minute: choose one organ you dissected and explain how its structure helps it do its job.' Allow WAIT TIME of 10-15 seconds, then say 'Share with your partner, then be ready to share with the class.' Cold-call 5 learners to explain one structure-function link; probe with 'What evidence from your specimen supports that?' and 'Can someone add to that?' Introduce the hormonal flowchart briefly, then ask groups to complete it on manila paper. Set debate ground rules: 'Respect every speaker, argue from evidence, no personal comments, and cultural or personal beliefs are respected while we focus on biology.' Give each side 3 minutes to prepare and 2 minutes to present, using the argument rubric (claim, evidence from dissection, reasoning, respectful rebuttal). Close by summarising: 'Frogs depend on numbers, mammals on protection and investment, and both achieve successful offspring.' Correct misconceptions, for example that IVF babies do not grow wholly in a test tube because fertilisation may happen in a dish but the embryo is placed in the uterus to develop.	Think-Pair-Share, Concept Mapping/Hormonal Flowcharts and Structured Debate: individual thinking, then peer talk and written maps help learners organise structure-function relationships. Debate makes them use evidence to justify a claim and compare the frog and cow strategies, which builds Engaging in Argument from Evidence. This is the Explain step of Predict-Observe-Explain, where they reconcile their predictions with observations.	Debate argument rubric (4 levels) scoring claim, evidence from the dissection, reasoning linking structure to function, and respectful rebuttal. Peer review of the concept map: partners check that at least 4 hormone-organ links are correct and give one 'star' and one 'step'. Observable indicators: learners correctly state that the uterus supports the embryo, the testes produce sperm, the ovaries produce eggs, and the oviduct is the site of fertilisation.
Driving Question	(8 min) Learners return to the class Driving Question Board. On	Say: 'Look at the board. Which questions have we answered	Driving Question Board review: making questions visible tracks	Quality of questions: look for at least 3 new questions that move

Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: reproduction animals mammalian reproductive  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals mammalian reproductive	sticky notes, each pair writes (a) one question they can now answer, with evidence from the dissection, and (b) one new question that arose, such as 'How do the sperm find the egg in the oviduct?', 'What happens when the oviducts are blocked?', 'How does IVF bypass the blocked tube?' or 'Why do cows produce one calf and rats many pups?' They place the notes on the board under categories: Answered, Partly Answered, New Questions.	today? Which new ones do you have?' Allow WAIT TIME of 10-15 seconds, then cold-call 3 pairs to read and place a note. Group similar questions, and point out the question that leads to the next lesson: 'We now know where the egg and sperm are made and where they might meet. But how do human sperm and egg cells meet and interact at the cellular level?' Link back to the phenomenon: 'How might this help explain why IVF is used when the oviducts are blocked?' Ask learners to take a note of the question for the next lesson.	growing understanding, shows learners that science is driven by their own questions, and creates a purposeful link between lessons.	from naming parts to asking about processes (for example fertilisation, implantation, IVF). Exit ticket (2 min): 'Name one structure and explain how its structure helps it work.' Check that 80% give a correct structure-function pair.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: reproduction animals mammalian reproductive  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms:	(12 min) In their groups, learners revise the class 'Reproduction Strategies' model started in Lesson 1 by adding a labelled diagram of the mammalian reproductive system (male and female) next to the frog's external fertilisation. They add arrows showing the gamete pathway: where the gametes are produced, where they travel, where fertilisation happens and where the embryo develops, and a note about where biotechnology like IVF could step in. They also record one remaining uncertainty.	Say: 'Using your drawings and table, add to our model. Show the path of the egg and sperm, and mark where things could go wrong.' Circulate and ask: 'What does your model now explain that it could not before?' 'Where might a blockage affect fertilisation?' Allow WAIT TIME of 10-15 seconds. Cold-call 3 groups to explain one revision. Ask 2 groups to present a model improvement to the class, and praise precise use of scientific terms. Hand out exit tickets. Collect and check for the common misconception that the whole embryo develops outside the body in IVF, and plan to address it in Lesson 3.	Cumulative Model Revision: adding mammalian structures to the earlier external fertilisation model lets learners compare strategies and see the model as a tool that explains the phenomenon. It also shows what remains unexplained, which motivates the next lesson.	Model checklist: (a) male and female structures correctly placed; (b) gamete pathway shown with arrows; (c) site of fertilisation and of embryo development identified; (d) link to frog strategy and one possible reproductive problem; (e) one unanswered question recorded. Exit ticket: 'Explain in two sentences why a cow can produce one calf at a time while a frog needs thousands of eggs.'

reproduction animals mammalian reproductive				
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D. TEACHER REFLECTION

After the lesson, reflect: Did all learners, including those uncomfortable with dissection, take part meaningfully through the different roles? Which structure-function links were hardest (for example the epididymis or the cervix)? Did learners use dissection evidence in the debate or only recall facts? Were any misconceptions (for example embryo growth in the stomach, IVF babies grow in test tubes) still present in the exit tickets? Was the language used respectful and scientific? What adjustments to the specimen or models, time and group size are needed for the next lesson, and are pre-downloaded animations ready for Lesson 3 in case of poor internet?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We dissected a small mammal and used models and charts to observe the testes, epididymis, sperm duct and urethra in males, and the ovaries, oviducts, uterus, cervix and vagina in females. We noted their shape, position and texture.
What did I learn?	Each part of the mammalian reproductive system is adapted to its function: the testes and ovaries produce gametes, tubes transport them, the oviduct is where fertilisation occurs, and the muscular, well-supplied uterus supports the developing embryo. Internal development protects a few offspring, unlike the frog's many eggs.
How does this explain the phenomenon?	A cow can give birth to one well-protected calf because its reproductive structures keep the gametes safe, bring them together inside the body and support the embryo for months, whereas frogs near Kisumu rely on many eggs because external fertilisation is risky. This also helps explain how science such as IVF can assist when part of the pathway fails, but we still need to find out how sperm and egg meet and interact at the cellular level.

LESSON 3 (80 minutes (2 x 40-minute periods)): Built for the Journey: Structure of Sperm and Ovum

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use drawings and microscope slides or photomicrographs to work out how the structures of sperm and ovum are specialised for movement, protection and fusion, and why this helps explain how frogs, hens, cows and humans achieve successful fertilisation.
Knowledge	By the end of the lesson the learner should be able to: relate the structure of the sperm cell and ovum to their functions; describe the process of fertilisation, including the role of the acrosome, flagellum, mitochondria, jelly coat/zona pellucida and haploid nuclei; and compare the numbers and sizes of gametes in frogs, hens, cows and humans.
Skills	SEP: Developing and using models (scale drawings and labelled diagrams); Constructing explanations (claim-evidence-reasoning linking structure to function); Analysing and interpreting data (comparing gamete size and number); Obtaining, evaluating and communicating information (peer presentations). Core competencies: Critical Thinking and Problem Solving; Learning to Learn.
Attitudes	Social Justice: learners share drawing, labelling and presenting roles fairly in groups. Peace: learners resolve differences amicably when peers disagree about a structure's function. Appreciation of the precision of living cells and respect for human sexuality topics treated scientifically (PCI: Human Sexuality: learners address myths and misconceptions).
Key Inquiry Question	How do animals reproduce?
Purpose in Storyline	Lessons 1-2 established that reproductive strategies differ (external fertilisation in frogs, internal fertilisation in hens and cows) but all succeed in bringing gametes together. Lesson 3 zooms to the cellular level to show how gametes are built for the job, and explains partly why frogs need thousands of eggs while cows produce one calf. It also addresses the IVF misconception: in IVF only fertilisation happens outside the body, and the embryo then develops in the mother or surrogate. It leads to: What controls the monthly release of these gametes and prepares the body for pregnancy?
Safety Notes	Handle microscopes and glass slides carefully; carry microscopes with two hands; discard broken slides in the sharps container. Use only prepared slides or photomicrographs, with no collection of any human samples. Treat all sexuality content in a neutral, scientific and respectful way, using correct anatomical terms; give learners the option to opt out of discussion without penalty. Keep any pre-downloaded clips on the teacher device or a shared device. Dissection from earlier lessons: wash hands with soap and clean surfaces before the lesson if the room was used for specimens.

B. LESSON OVERVIEW

Learners start by recalling the phenomenon: frogs near Kisumu shedding thousands of eggs into Lake Victoria wetlands, hens laying shelled eggs, cows giving birth to single calves and IVF stories in the news. They predict how a sperm and an ovum differ in structure and what each cell needs to achieve fertilisation. In groups sharing microscopes, textbooks or photomicrograph charts, they observe prepared slides of sperm (and of mammalian ovum or follicle if available), then draw and label each gamete with scale notes. Using Think-Pair-Share and claim-evidence-reasoning, they explain how each structure suits movement, protection or fusion, then present to peers using a gallery walk and peer review. Learners add new questions to the Driving Question Board, notably what controls the monthly release of gametes, and revise their class model by adding the cellular level of fertilisation, correcting the IVF misconception.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction animals gamete sperm	Learners individually look at the phenomenon image strip (frog eggs in a Kisumu pond, a hen's egg, a calf, an IVF news headline) and answer in their notebooks: 'A sperm cell and an ovum must meet and fuse. What features would each cell need to make this happen? Sketch your prediction of each cell and give one reason for each feature.' They then compare sketches with a partner and note one thing they agree on and one they disagree on. Learners also predict: 'Which is bigger, and why might that be?'	Display the phenomenon images. Say: 'Remember the frogs at the Lake Victoria wetlands and the cow with one calf. Before we look at any slides, I want your best thinking. There are no wrong predictions today.' Pose: 'If you were designing a cell to swim to another cell, what would you give it? What would you give the cell that waits?' Give 2 minutes of silent sketching, then WAIT TIME of 10-15 seconds before calling on anyone. Cold-call 4 learners (two for sperm ideas, two for ovum ideas) and record the predicted features on the board in two columns without evaluating them. Ask: 'Who has a different idea? What made you think that?' Circulate and photograph or note 3 interesting predictions to revisit later.	Predict-Observe-Explain (Predict stage): Committing to a prediction first makes learners' prior ideas visible and creates a reason to look closely at the slides. Think-Pair-Share lets every learner voice an idea before whole-class talk, which builds a bridge from the animals' visible reproductive strategies to the invisible cellular structures.	Check notebook sketches for observable indicators: (1) a tail or other means of movement drawn on the sperm; (2) a larger cell with food reserves or protective layer drawn for the ovum; (3) a reason written beside each feature. Tally on a class tally sheet how many learners predicted movement for sperm and stored food/protection for ovum. Note misconceptions such as 'sperm and ovum are the same size' or 'the ovum swims'.
Observe Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction	In groups of 4 sharing microscopes, textbooks or printed photomicrographs, learners rotate through roles (microscope operator, drawer, labeller, recorder). They observe high-power prepared slides or photomicrographs of mammalian sperm (e.g. bull or human) and a mammalian ovum/follicle, plus a pre-downloaded or printed labelled comparison chart of frog egg, hen's egg yolk and human ovum. They make large, pencil-only drawings of one sperm and one ovum with title, magnification note and label lines, and add a table recording size, shape, movement, food reserves and protective layers. They then compare gamete numbers: frog eggs in a spawn mass (thousands), hen (about one	Hand out slides and charts and model the drawing conventions on the board (clear lines, no shading, label lines ruled and not crossing). Say: 'Look first, draw second. Describe to your group what you see before you write a label.' Explain that sperm are about 50 micrometres long and the human ovum about 0.1 mm, just visible as a dot. Circulate and ask each group: 'Which part of this cell would help it move? How do you know from the structure?' and 'Where do you see stored food or protection?' WAIT TIME of 10-15 seconds after each question. Ensure fair role rotation by calling 'Switch roles now' at the halfway point (Social Justice). Remind learners of microscope safety. For groups with no working	Predict-Observe-Explain (Observe stage): Direct observation of real cells and scale drawings turns abstract textbook claims into evidence learners generated. Drawing forces attention to features such as the midpiece packed with mitochondria. Comparing gamete sizes and numbers across frog, hen, cow and human links the cellular evidence back to the anchoring phenomenon.	Use an annotated-drawing checklist: sperm head with nucleus and acrosome, midpiece with mitochondria, and tail/flagellum labelled; ovum with nucleus, cytoplasm with food reserves, and jelly coat/zona pellucida labelled; title, magnification and ruled label lines present. Observation table complete with at least 4 comparisons. Oral probe indicators: learner can point to a structure and state one function using the word 'because'.

animals gamete sperm	egg a day), cow (usually one calf), and human (usually one ovum per cycle).	microscope, give priority to the photomicrograph and the chart. Cold-call 3 groups to report one surprising observation.		
Explain Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction animals gamete sperm	Learners complete a claim-evidence-reasoning (CER) frame for each gamete: 'The sperm is adapted for movement and fusion because...' and 'The ovum is adapted for protection and nourishment because...'. They explain the functions of the acrosome (enzymes that digest the jelly coat/zona pellucida), the flagellum, mitochondria (energy), the haploid nucleus of each gamete, and the ovum's food reserves and protective layer. They then use Think-Pair-Share to answer: 'Why do frogs need thousands of eggs but cows only one calf?' using evidence on external vs internal fertilisation, protection and parental care. Groups present their drawings in a gallery walk, giving each other 'two stars and a wish'. Finally learners correct the IVF idea: 'Babies do not grow in a test tube; only fertilisation happens in the laboratory dish, and the embryo is later placed in the womb.'	Say: 'Use your drawings as evidence. Every claim needs a structure you actually saw.' Give 3 minutes of silent writing for CER, then pair talk for 2 minutes. WAIT TIME of 10-15 seconds before calling on learners. Cold-call 5 learners for different structure-function links (acrosome, tail, mitochondria, jelly coat, food reserves) and ask a follow-up: 'What evidence in the drawing supports that?' Pose the numbers question: 'In Lake Victoria, a frog releases eggs into open water. What threatens them? What does a cow's body provide?' Show a short pre-downloaded animation clip of fertilisation (or a printed sequence) for 3 minutes, then ask: 'What did the clip show that the slide could not?' Run the gallery walk for 8 minutes with a 'switch' signal. Address the IVF misconception directly with a diagram: 'Where does fertilisation happen in IVF? Where does the baby grow?' Keep the tone neutral and scientific, using proper anatomical terms, and redirect any joking calmly: 'We use correct scientific words and respect each other.' Resolve disagreements: 'State your evidence, then listen to your peer's evidence' (Peace).	Claim-Evidence-Reasoning with Think-Pair-Share: CER requires learners to tie each functional claim to a structure they drew, building causal explanations rather than memorised labels. Pair talk rehearses scientific language before public sharing. The gallery walk and peer review broaden the evidence base and let learners correct each other's diagrams. Linking to frogs versus cows makes structure-function reasoning explain the anchoring phenomenon.	Observable indicators: CER statements name a structure, state its function and link them with 'because'; at least 3 of 5 structure-function pairs correct; learner explains that fewer eggs are needed when fertilisation is internal and offspring are protected; learner states that IVF fertilisation is in a dish and development occurs in the womb. Peer review of drawings using the checklist (labels correct, ruled lines, title, scale). Teacher records common errors (e.g. acrosome labelled as tail) for next-day reteaching.
Driving Question Board (DQB) Creation  VIDEO: Fertilization terminology:	Learners return to the class DQB. They write on sticky notes (a) one question that is now answered, with the evidence, and (b) one new question that today's observations raised. Likely questions: 'How is the ovum released each month?',	Say: 'Look at our Driving Question Board. Which questions can we now answer using what we saw today? Move them to the Answered column and tell us your evidence.' Give 3 minutes of writing, then WAIT TIME of 10-15	Driving Question Board review: Moving questions between 'unanswered' and 'answered' with cited evidence makes progress in understanding visible and gives learners ownership of the next steps. New questions about timing	Observable indicators: at least one note per learner with a question stem that is investigable (starts with how/what controls/why); an 'answered' note includes cited evidence from slides or drawings; at least 3 learners ask questions

gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: reproduction animals gamete sperm	'Why only one ovum in humans?', 'What happens if two sperm reach the ovum?', 'What controls the timing?'. Learners place notes under categories (Gametes, Fertilisation, Hormones, Pregnancy, Biotechnology), and the class chooses one question to investigate next.	seconds before sharing. Cold-call 4 learners to read and place a question and ask 'What made you wonder that?'. Highlight the question: 'What controls the monthly release of these gametes and prepares the body for pregnancy?' and say: 'This is where we go next.' Group similar questions with the class and ask: 'Which questions are about hormones? Which are about development?' Keep sensitive questions respectful; where a question is not appropriate for class, collect it on a private question box and answer it neutrally in general terms next lesson.	and control create a genuine need for the next lesson on hormones and the menstrual cycle.	that link to hormones or timing. Collect exit tickets: 'One structure of sperm or ovum and how it helps fertilisation; one question I still have.'
Model Building Phase VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: reproduction animals gamete sperm	Learners revise the class model of 'How do animals ensure successful fertilisation and development of offspring?' In groups they add a new zoomed-in panel showing a sperm approaching an ovum, with labelled adaptations (acrosome, flagellum, mitochondria, jelly coat, haploid nuclei), and link it to the earlier external and internal fertilisation panels. They add an arrow marked 'IVF: fertilisation in dish, embryo then placed in womb' and a note comparing gamete numbers in frogs and cows. Each group adds a question mark box: 'What controls when the ovum is released?' to show what they cannot yet explain. They complete the summary table below in their notebooks.	Say: 'Our model now needs to go inside the cells. Add what you learned, and show how it explains the frogs, hens, cows and IVF.' Provide manila paper or reuse the class model chart. Give 8 minutes for building and 2 minutes for a gallery review. Ask each group: 'Which part of your model explains why a frog needs thousands of eggs?' and WAIT TIME of 10-15 seconds. Cold-call 3 groups to explain one change they made and why. Prompt: 'What does the model still not explain?' and record gaps for Lesson 4. Collect exit tickets and drawings for marking.	Cumulative model revision (concept map linking structure, function and strategy): Adding the cellular level to the whole-organism model shows how structure explains function across scales and keeps the gaps visible. Identifying what the model cannot yet explain (timing of ovum release) builds motivation for the next lesson.	Observable indicators: model includes at least 4 labelled gamete adaptations with functions; shows the link between external/internal fertilisation and gamete number; correctly shows IVF with development in the womb; contains at least one unanswered question about timing or control. Exit ticket scored 0-2 (0 no correct structure-function link, 1 one, 2 two or more with reasoning).

D. TEACHER REFLECTION

Which structure-function links did most learners explain with evidence, and which (for example the acrosome or mitochondria) needed reteaching? Did groups with limited microscope access still gather good evidence from photomicrographs? Did every learner take a turn drawing and presenting (Social Justice)? Was the IVF misconception corrected, and did the discussion stay respectful and scientific? Which DQB questions about hormones should open Lesson 4?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What did you observe about the structure of the sperm cell and ovum (shape, size, tail, acrosome, mitochondria, jelly coat, food reserves) and about the number of gametes in frogs, hens, cows and humans?
What did I learn?	What did you learn about how each structure helps movement, protection and fusion during fertilisation, and about what happens in IVF compared with natural fertilisation?
How does this explain the phenomenon?	How can you use the structure of sperm and ovum to explain why frogs near Lake Victoria need thousands of eggs while cows produce one calf, and what do you still wonder about how the body controls the monthly release of the ovum?

LESSON 4 (80 minutes (double lesson)): Hormonal Conductors: How Pituitary and Ovarian Hormones Regulate the Menstrual Cycle and Egg Release

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use hormone charts and graph worksheets as evidence to figure out how pituitary and ovarian hormones interact through feedback mechanisms to regulate the human menstrual cycle and gamete release. This helps explain why a cow or a woman releases one egg at a time, in contrast to the thousands of eggs a frog releases, and how science can help when this regulation fails.
Knowledge	Describe the role of hormones (FSH, LH, oestrogen, progesterone) in the human menstrual cycle. Explain how negative and positive feedback between the pituitary gland and the ovaries controls follicle development, ovulation, and preparation of the uterine lining. Predict the effects of undersecretion and oversecretion (imbalance) of reproductive hormones.
Skills	Developing and using models (hormonal flowcharts/concept maps); analysing and interpreting data (reading hormone graphs); constructing explanations from evidence; engaging in argument from evidence (structured debate); communicating with peers in groups sharing limited textbooks and devices.
Attitudes	Core competencies: Critical Thinking and Problem Solving (giving objective ideas to solve reproductive health problems); Learning to Learn (asking and seeking information from a resource person). Values: Social Justice (sharing group roles fairly); Peace (resolving differences amicably in debate). PCIs: Menstrual Health Disorders Management (sensitisation on human menstrual health while illustrating the role of hormones in the human sexual cycle); Human Sexuality (understanding common myths and misconceptions around human sexuality while illustrating the role of hormones in the human sexual cycle).
Key Inquiry Question	How do animals reproduce?
Purpose in Storyline	Lessons 1-3 showed that animals use different fertilisation and development strategies (frogs externally, hens and cows internally) and examined reproductive structures. Lesson 4 gives the internal control system: hormones decide when and how many gametes are released. It builds on the reproductive system structures and the idea that cows and humans release few eggs, and it sets up Lesson 5: How does the hormonal balance shift if fertilisation occurs and leads to pregnancy? It also lays the groundwork for understanding why IVF uses hormone treatment when natural regulation fails.
Safety Notes	Human reproduction topics are sensitive. Set ground rules at the start: respectful language, no personal questions, no pressure to share personal experiences, and use of the anonymous question box. Use scientific terms and diagrams only. Address myths factually and without ridicule (e.g. that menstruation makes a person unclean or unable to cook, touch crops or attend school). Be aware of learners who may be uncomfortable and allow them to pass. Offer a private route for health concerns, such as the school nurse or guidance and counselling teacher. If the small mammal dissection is revisited as a POE link (for example, ovaries and uterus of a rat), follow school safety standards and animal welfare guidelines: use only specimens obtained ethically, wear gloves, lab coats and masks, keep sharp tools under teacher control, disinfect surfaces, wash hands, and dispose of specimens as directed. Debates must remain neutral and scientifically grounded and respect diverse cultural and religious backgrounds.

B. LESSON OVERVIEW

Learners return to the anchoring phenomenon: frogs release thousands of eggs, a hen lays one egg at a time, a cow produces one calf, and women usually release one egg per month. They ask what controls this timing and number, and why IVF often needs hormone treatment. In a Predict-Observe-Explain sequence, learners first predict how hormone levels change over a 28-day cycle, then trace FSH, LH, oestrogen and progesterone on pre-printed charts and graph worksheets (no internet

required). They then explain the patterns using feedback mechanisms, build a hormonal flowchart/concept map, and take part in a structured, respectful debate on myths, imbalances and menstrual health. The Driving Question Board (DQB) is reviewed and new questions are added, leading to the key question for Lesson 5: how does the hormonal balance shift if fertilisation occurs? Learners revise their cumulative model by adding a hormone control layer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Hormone concentration metabolism and negative feedback Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Menstrual Cycle Practice Source: CK-12  Search ARES for similar readings	Learners recall the phenomenon: a frog pond near Kisumu with thousands of eggs, a hen laying one egg every day or so, a cow releasing one egg per cycle, and a news story about a Nairobi couple using IVF with hormone injections. Individually, then in pairs (Think-Pair-Share), they sketch on a blank 28-day axis how they think the levels of FSH, LH, oestrogen and progesterone change, marking the day they think ovulation occurs. They write one sentence: 'I think hormones control egg release by...' and note one question or myth they have heard about the menstrual cycle on the anonymous question slip.	Open with: 'Last lesson we looked at the structures. Today we ask: what tells an ovary WHEN to release an egg, and why only one?' Display the ground rules and read them aloud. Say: 'In your notebook, sketch what you think happens to four hormones over 28 days. There is no wrong prediction today; it is your starting model.' Give 3 minutes of silent individual work. WAIT TIME: 10-15 seconds after asking 'Which hormone do you think peaks just before ovulation, and why?' before taking any responses. Then Pair (2 minutes) and Share: cold-call 4 pairs (not volunteers), recording their predictions on the board in a 'Predictions' column without evaluating. Collect the anonymous slips in a box. Probe: 'What in your experience or in the frog, hen and cow examples makes you think that?' Circulate with a checklist noting prediction types: correct sequence, reversed, or unsure.	Think-Pair-Share / Predict phase of Predict-Observe-Explain: committing to a prediction surfaces prior ideas and misconceptions (e.g. that only one hormone is involved, or that ovulation occurs at menstruation) so learners have a reason to look closely at the evidence in the next phase.	Observable indicators: (1) the sketch includes all four hormones with labels; (2) the learner marks a day for ovulation, with a reason given; (3) pairs justify the prediction using a reason, not a guess; (4) the teacher tallies on a checklist how many learners place the LH peak before ovulation, and how many show progesterone rising in the second half of the cycle. The tally informs the pacing of the Observe phase.
Observe Phase  VIDEO: Hormone concentration metabolism and negative feedback Source: Khan Academy (English - US curriculum)	In mixed-ability groups of 4 (sharing one chart set and textbook), learners use the menstrual cycle hormonal charts and graph worksheets to trace the four hormones across Days 1-28. Roles are rotated fairly (Social Justice): Reader, Tracer (uses coloured pencils: FSH-blue, LH-green, oestrogen-red,	Distribute chart sets and say: 'You are data detectives. Your job is to describe only what the graph shows today; explanations come next.' Model tracing for Days 1-5 on the board. Circulate with these probes: 'Which hormone is rising on Day 10? What is happening in the ovary at that time?' and 'Point to where the LH peak sits relative	Predict-Observe-Explain (Observe): learners collect evidence from data representations, comparing patterns with their predictions. Graph tracing builds data-literacy and reveals patterns (timing, order, peaks) that become the evidence base for explanation. Rotating roles builds fair sharing of	Observable indicators: completed observation tables with correct sequence of hormone peaks (FSH early, oestrogen before LH, LH peak before ovulation, progesterone after ovulation); accurate use of colours and labels on traced graphs; learners use terms such as 'peak', 'surge' and 'falls' correctly in oral reports;

Search ARES for similar videos EXERCISE: Menstrual Cycle Practice Source: CK-12 Search ARES for similar readings	progesterone-orange), Recorder, and Reporter. They complete an observation table: Day range, hormone levels (rising, peak, falling), ovary event (follicle growth, ovulation, corpus luteum), uterine lining event (menstruation, repair, thickening, maintained/breakdown). They identify: the oestrogen rise before the LH surge; the LH peak around Day 14 and ovulation soon after; the progesterone rise after ovulation; the fall of both before menstruation. Learners then examine three 'imbalance' scenario cards (e.g. low FSH, excess oestrogen/irregular cycles, absent LH surge with no ovulation) and annotate how each graph differs from the normal one, connecting this to why IVF patients may receive FSH injections to stimulate several follicles, unlike the one egg normally released. Pre-downloaded animation clips of the cycle may be played on one shared device per group or projected for the class.	to the oestrogen peak.' WAIT TIME: 10-15 seconds after each probe before prompting. After 15 minutes, cold-call 3 groups to report one pattern each. Ask: 'What did you notice that surprised you compared with your prediction?' Distribute imbalance cards and ask: 'Compare this graph with the normal one. What is missing or different, and what might the effect on egg release be?' Remind learners that these are scientific scenarios, not diagnoses of anyone in the room. Use the pre-downloaded animation (about 3 minutes) for groups that need visual support. Ensure every learner has a turn with the chart, so that sharing of devices and books is equitable.	responsibility.	teacher records for each group whether they identified the LH surge and the post-ovulation progesterone rise; peer check: groups swap charts and tick or correct two items.
Explain Phase VIDEO: Hormone concentration metabolism and negative feedback Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Menstrual Cycle Practice Source: CK-12 Search ARES for similar readings	Learners move from 'what' to 'why'. In groups, they build a hormonal flowchart/concept map linking: hypothalamus/pituitary, FSH, follicle growth, oestrogen, uterine lining thickening, LH surge, ovulation, corpus luteum, progesterone, and menstruation. They add arrows labelled as negative feedback (e.g. rising progesterone and oestrogen inhibit FSH and LH release) or positive feedback (high oestrogen triggers the LH surge). They then use their map to explain, in 3-4 sentences, why only one egg is normally released each cycle and how this differs from the frog's thousands of	Say: 'Now we move from what happened to why it happened. Use your charts as evidence for every arrow you draw.' Demonstrate one feedback arrow on the board: 'High progesterone tells the pituitary to reduce FSH and LH. This is negative feedback, like a thermostat.' WAIT TIME: 10-15 seconds after asking 'Why do you think oestrogen can switch from inhibiting LH to triggering a surge?' Cold-call 3 groups to present one link on their map, and ask the others: 'Do you agree or disagree, and what evidence from the graph supports your view?' For the debate, assign sides randomly,	Concept Mapping/Hormonal Flowcharts and Structured Debate: concept maps force learners to show relationships and feedback loops, not just list facts, making causal reasoning visible. Structured debate requires learners to use evidence to argue, rebut and resolve differences peacefully (Peace value), and it exposes misconceptions so that they can be corrected with science.	Observable indicators: flowchart includes at least 8 correct linked terms and correctly labelled negative/positive feedback arrows; written explanation states that LH surge triggers ovulation and that progesterone maintains the lining; debate rubric scores (1-4) for use of evidence, scientific accuracy, respectful language and responsiveness; learners correct at least one myth with a scientific statement; teacher notes learners who still confuse oestrogen and progesterone roles for reteaching.

	eggs, then predict what might happen if one hormone is too low or too high (e.g. no LH surge means no ovulation; this is where fertility treatment may help). A structured debate follows with two sides on the motion: 'Understanding hormones helps remove myths and improve menstrual health in our communities (such as in schools in Kisumu or Kericho).' Side A argues the value of open, scientifically accurate education; Side B argues from the perspective of cultural sensitivity and the need for respectful approaches, with both sides required to use evidence from the charts. Learners discuss myths, for example that menstruation is a sickness, that a girl cannot touch food during her period, or that irregular cycles always mean infertility, and replace them with scientifically accurate statements. A short discussion covers when to seek help (for example very painful periods, very irregular cycles, or absent periods) through the school nurse or a health facility, as part of Menstrual Health Disorders Management.	give 4 minutes preparation, and use a rubric (accuracy of science, use of evidence, respectful language, responding to the other side). State: 'We argue the ideas, never people. Use evidence from the charts.' Intervene calmly if language becomes disrespectful and restate the ground rules. After the debate, ask a neutral summary question: 'What did both sides agree on scientifically?' and cold-call 3 learners. Address myths factually: 'What does the evidence on the graph show about any link between menstruation and being unclean?' Close with one key message: 'Hormones coordinate the timing, and understanding them protects health.' Invite the anonymous question slips and answer 2-3 scientifically, deferring others to the school nurse or counselling teacher if they are personal.		
Driving Question Board (DQB) Creation  VIDEO: Hormone concentration metabolism and negative feedback Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners return to the class DQB. Individually, they write on sticky notes: (1) one question from today that has been answered (placed in the 'Answered' area with evidence from the charts); (2) one new question that has come up (for example, 'Why does the cycle stop during pregnancy?', 'What happens to the hormones if the egg is fertilised?', 'How do IVF clinics in Nairobi use hormones?', 'How do cows and goats on Kenyan farms control their	Say: 'Look at the DQB. Which questions can we now answer with evidence? Which new questions did the data raise?' Give 3 minutes for individual note-writing. WAIT TIME: 10-15 seconds after asking 'Which question on the board connects to what we saw on the graph today?' Cold-call 4 learners to place and justify a question. Move the answered questions and star the new question 'How does the hormonal balance shift if fertilisation occurs and leads to	Driving Question Board (DQB) review: tracking answered and unanswered questions shows learners how their understanding grows, gives them ownership of the inquiry, and creates a purposeful link from the evidence in Lesson 4 to the investigation in Lesson 5.	Observable indicators: exit ticket correctly names LH (surge) as the trigger for ovulation and gives a reasonable imbalance effect (e.g. low FSH leads to poor follicle development); sticky-note questions are testable and linked to the evidence rather than recall only; learners can justify the placement of a question as 'Answered' using chart evidence; teacher tallies the proportion of exit tickets with correct answers to decide on a brief review at the

 EXERCISE: Menstrual Cycle Practice Source: CK-12  Search ARES for similar readings	cycles?'). In pairs, they group questions into themes (feedback, imbalance, pregnancy, biotechnology). The class reviews the question 'How does the hormonal balance shift if fertilisation occurs and leads to pregnancy?' and identifies it as the next lesson's focus. Learners also complete an exit ticket: 'Name the hormone that triggers ovulation and explain one way a hormone imbalance could affect the cycle.'	pregnancy?' to the 'Next Lesson' area. Link back to the anchoring phenomenon: 'Remember the frog with thousands of eggs and the cow with one calf: which questions on our board still need answers about development and pregnancy?' Collect exit tickets at the door and read them before the next lesson.		start of Lesson 5.
Model Building Phase  VIDEO: Hormone concentration metabolism and negative feedback Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Menstrual Cycle Practice Source: CK-12  Search ARES for similar readings	Learners revise their cumulative class/individual model of how animals ensure successful fertilisation and development of offspring. In their notebooks, they add a new layer: 'Hormonal control of gamete release', including a simplified diagram of the pituitary-ovary feedback loop and a cycle graph with the four hormones. They annotate links to previous lessons: external fertilisation in frogs (large numbers of eggs, environmental cues) versus internal fertilisation in cows and humans (timed release of one or few eggs regulated by hormones). They add a note on how biotechnology, such as IVF, can use hormone treatment to help when natural regulation fails, and correct the misconception that IVF babies develop wholly in a test tube (only fertilisation occurs in the laboratory, and development takes place in the uterus). Learners leave a 'gap' box where they predict what will change in the model if fertilisation occurs (Lesson 5).	Say: 'Open your model from Lesson 3. What does it need to explain today's evidence? Add the hormone layer and show where feedback fits.' Provide a sentence frame: 'Because ____ (hormone) increases, ____ happens, which leads to ____.' WAIT TIME: 10-15 seconds after asking 'What part of your model explains why a woman usually releases one egg while a frog releases thousands?' Cold-call 3 learners to share a revision, and invite one to explain what they changed and why. Ask: 'What evidence made you change your model?' Circulate and check that models include correct arrows, labelled hormones and a link to the phenomenon. Close by revisiting the IVF misconception: 'Where does the baby actually develop in an IVF pregnancy, and what role do hormones play?' End with the Lesson 5 hook: 'If an egg is fertilised, what do you predict will happen to progesterone, and why might the cycle stop?'	Cumulative model revision (concept mapping): updating a model with new evidence shows how scientific explanations are refined, and linking hormones to the anchoring phenomenon helps learners integrate cellular, organ-level and technological ideas into a coherent explanation.	Observable indicators: the revised model contains a labelled feedback loop with at least FSH, LH, oestrogen and progesterone; annotations link hormone control to a comparison between frog and cow/human egg release; learner explicitly corrects the IVF 'test tube baby' misconception; the learner's gap prediction for Lesson 5 is reasonable (e.g. progesterone remains high); teacher uses a 3-point check (complete, partial, missing) for each element and gives written feedback on the models.

D. TEACHER REFLECTION

After the lesson, consider: Did learners trace all four hormones accurately and distinguish oestrogen from progesterone? Which feedback ideas (negative versus positive) remained confusing, and what visual or analogy (e.g. a thermostat or a tap) helped? Did the ground rules and anonymous question box create a safe, respectful space, and were myths corrected without embarrassing anyone? Were roles shared fairly in groups given limited devices and textbooks? Review exit tickets to decide what to revisit at the start of Lesson 5. Note any learners who need follow-up through the school nurse or guidance and counselling teacher. If the earlier dissection links were used, confirm that safety and animal welfare standards were followed. Record how well learners connected hormonal control to the frog, hen and cow phenomenon.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Hormone charts and graphs showed FSH rising early in the cycle, oestrogen rising before a sharp LH surge near Day 14, ovulation shortly after the LH peak, and progesterone rising in the second half of the cycle, then falling before menstruation. Imbalance graphs differed from the normal pattern.
What did I learn?	The pituitary gland (FSH and LH) and the ovaries (oestrogen and progesterone) interact through negative and positive feedback to time follicle development, ovulation and preparation of the uterine lining. Hormonal imbalances can disrupt ovulation and the cycle, and accurate knowledge helps to dispel myths and manage menstrual health.
How does this explain the phenomenon?	Humans and cows usually release one egg at a time because hormone feedback loops control when a single follicle matures and ovulates, whereas frogs rely on large numbers of eggs released into water. When natural hormone regulation fails, biotechnology such as IVF can use hormone treatment to assist, but the baby still develops in the uterus. This leads to the next question: how does the hormonal balance shift if fertilisation occurs?

LESSON 5 (80 minutes (double lesson)): From Fertilisation to Birth: How Hormones and the Placenta Sustain Pregnancy

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence from pre-downloaded animations and a nurse/midwife about how fertilisation, implantation, placenta formation, hormonal control of pregnancy and birth ensure development of offspring in humans. They compare this with the cow's live birth from the anchoring phenomenon.
Knowledge	By the end of the lesson, the learner should be able to: (d) describe the process of fertilisation, pregnancy and birth in humans; (e) describe the role of hormones in human reproduction (hCG, progesterone, oestrogen, oxytocin, prolactin); explain how the placenta exchanges nutrients, gases and wastes without mixing maternal and fetal blood.
Skills	Developing and using models (hormonal flowcharts and concept maps); obtaining, evaluating and communicating information (animations, guest speaker); constructing explanations using evidence; engaging in argument from evidence (structured debate); drawing and labelling.
Attitudes	Appreciate the role of skilled antenatal and delivery care in maternal and child health. Show respect and maturity when discussing human sexuality, and respect diverse cultural views. Core competencies: Learning to Learn (seeking information from a resource person on stages of pregnancy and childbirth) and Critical Thinking and Problem Solving. Values: Peace (resolving differences amicably during debate) and Social Justice (sharing roles fairly in groups). PCIs: Human Sexuality (understanding common myths and misconceptions) and Menstrual Health Disorders Management.
Key Inquiry Question	How do animals reproduce?
Purpose in Storyline	Lessons 1-4 established external fertilisation (frogs), egg-laying (hens) and internal fertilisation types. This lesson explains how the cow-like strategy of live birth works in humans: hormones and the placenta support one well-nourished baby. It closes with a new question: What happens when natural fertilisation or pregnancy cannot occur? This leads to IVF, surrogacy and sperm banks in Lesson 6.
Safety Notes	Pre-download all animation clips (no internet is needed in class). Use only diagrammatic, age-appropriate animations. Agree on ground rules before starting: respectful language, no personal questions, and no jokes about bodies. Learners may pass or step out quietly. Guest speaker briefs learners on scientific content only. Provide a question box for anonymous questions. Any dissection or specimen revisit follows school safety standards: gloves, aprons, hand washing, no eating, disposal as per school protocol, and animal welfare guidelines. Devices are shared in groups of 4-5 and kept on low brightness to save battery.

B. LESSON OVERVIEW

Return to the phenomenon: a cow gives birth to one live calf after months of internal development, while frogs release thousands of eggs. Learners predict how a baby gets food, oxygen and protection inside the mother and what triggers birth. They watch pre-downloaded animations of fertilisation, implantation, placenta formation and birth, and question a nurse/midwife (for example from the local sub-county hospital). They use Think-Pair-Share, a hormonal flowchart, and a short structured debate on shared reproductive health choices to explain how hormonal shifts and mother-fetal structures work together. Learners update the Driving Question Board and revise their class model, then identify the gap: what if fertilisation or pregnancy cannot happen naturally?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction animals fertilisation implantation  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: reproduction animals fertilisation implantation	Learners recall the anchoring phenomenon: one calf born alive on a Kericho dairy farm versus thousands of frog eggs at a Kisumu pond. Individually, they sketch for 3 minutes how they think a baby inside a mother gets food and oxygen, and what starts birth. They add a prediction in a sentence: 'I think ____ because ____.' In groups of 4, they compare sketches and agree on one prediction and one question.	Display the question: 'A cow carries one calf for about 9 months. How does that calf get food and oxygen, and what tells the cow's body it is time to give birth?' Say: 'Draw first, talk second. There are no wrong predictions today; I want your current thinking.' Give WAIT TIME of 10-15 seconds after the question before anyone writes. Circulate and note misconceptions (e.g. 'the baby eats through the mouth', 'mother's blood flows directly into baby', 'IVF babies grow in test tubes'). Cold-call 4 groups: 'Group 3, what is your prediction and what is your evidence?' Record predictions on the board without evaluating. Remind learners of the ground rules for respectful discussion.	Predict-Observe-Explain (Predict step): Committing to a prediction in writing exposes prior ideas and makes learners notice when animation evidence confirms or contradicts them, which builds conceptual change. Think-Pair-Share (individual sketch, then pair/group comparison) ensures every learner voices a view before whole-class talk.	Check sketches for: (1) any arrow showing food/oxygen reaching the baby, (2) any trigger for birth named (hormone, pressure, signal), (3) misconceptions recorded on a tally sheet (e.g. direct blood mixing, feeding by mouth). Observable indicator: at least 80% of learners write a prediction with a reason. Tally which misconceptions occur most to target in the Explain phase.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction animals fertilisation implantation  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms:	In groups of 4-5 sharing a device, learners watch four pre-downloaded animation clips (about 3 minutes each): (1) fertilisation in the oviduct, (2) implantation and early hCG signalling, (3) placenta and umbilical cord formation and exchange, (4) labour and birth. For each clip they complete an observation table: structure seen, what moves/changes, hormone involved. Then the nurse/midwife speaks for 10 minutes on antenatal clinic visits, nutrition (e.g. iron-rich sukuma wiki, beans, omena), stages of labour and postnatal care. Learners ask prepared and anonymous-box questions, recording answers in their tables. They draw and label the placenta and umbilical cord,	Before playing: 'While you watch, focus on three things: what structure forms, what substance moves, and which hormone changes.' Pause each clip at key moments and ask: 'What do you see happening right now?' with WAIT TIME of 10-15 seconds. Replay the placenta exchange clip once. Rotate roles in groups (device operator, recorder, timekeeper, reporter) to share responsibility fairly. Brief the guest beforehand on the content scope and ground rules. Moderate Q&A: 'Please ask about the biology of pregnancy and care.' Cold-call 5 learners for questions across the groups, ensuring at least 2 come from quieter learners. Ask the guest probing questions: 'Why is it	Predict-Observe-Explain (Observe step): Structured observation tables focus attention on mechanism (structure, substance, hormone) rather than just watching. Consulting a resource person provides authentic, local evidence about real pregnancies in Kenya and strengthens Learning to Learn by practising how to question an expert.	Oral probes during guest Q&A: can learners ask a question that links structure to function (e.g. 'Why does maternal blood not mix with fetal blood?')? Check observation tables for the entries: fertilisation in oviduct; implantation in the uterine wall; placenta as site of exchange; umbilical cord carries blood; oxytocin in labour. Annotated drawing criteria: placenta, umbilical cord, amnion/amniotic fluid and uterus labelled correctly with at least 4 of 5 labels, and arrows showing the direction of oxygen, nutrients and wastes.

reproduction animals fertilisation implantation	indicating where exchange occurs.	important to attend antenatal clinics early?' Pass anonymous box questions to the guest and answer sensitively. Where a device fails, use printed charts or the textbook diagram.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: reproduction animals fertilisation implantation  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals fertilisation implantation	Learners individually write a 2-sentence explanation of how the baby gets food and oxygen. In pairs they build a hormonal flowchart showing: fertilisation, then implantation, then hCG keeps the corpus luteum active, then progesterone maintains the uterine lining and the placenta takes over hormone production, then late pregnancy oestrogen rises, then oxytocin triggers contractions (positive feedback), then birth, then prolactin and oxytocin support lactation. They compare the human placenta with the cow's placenta and with the hen's egg (yolk and shell) and frog's jelly eggs, explaining why one well-supported offspring needs fewer eggs. Finally they correct their earlier predictions in a different colour.	Say: 'Use evidence from the animations and from our guest to explain, not just what you remember.' Give WAIT TIME of 10-15 seconds before pairs share. Provide sentence frames: 'The placenta allows ___ because ___.' Cold-call 6 pairs to share one link on the flowchart; ask: 'What evidence from the animation supports that arrow?' and 'Which hormone is the trigger, and why does this count as positive feedback?' Address misconceptions identified earlier: blood does not mix; exchange occurs by diffusion across thin membranes; IVF embryos are placed in the uterus and still develop inside the mother. Ask: 'Why does a frog need thousands of eggs while a cow needs one calf?' and link to parental care and protection. Correct myths sensitively using scientific language (e.g. pregnancy depends on hormones, not on traditional beliefs about food or behaviours).	Concept Mapping/Hormonal Flowcharts: Sequencing hormone events and structures as arrows with causes forces learners to show mechanism and feedback, not just list facts. Think-Pair-Share lets learners rehearse explanations before presenting. Comparing the cow, hen and frog links the lesson back to the anchoring phenomenon and shows trade-offs between number of offspring and parental investment.	Flowchart rubric: (1) correct order of at least 6 events, (2) hormone named at each step, (3) placenta role included, (4) positive-feedback loop for oxytocin shown. Look for explanations that include 'diffusion', 'thin membrane', 'separate blood supplies'. Observable indicator: at least 70% of pairs correctly link hCG to maintaining progesterone. Exit ticket question (collect at end): 'Explain in 3 sentences why the placenta is called a lifeline, and name 1 hormone that starts labour.'
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: reproduction	Learners return to the class Driving Question Board. Groups write new questions on sticky notes, e.g. 'Why do some women need a caesarean section?', 'What if the egg and sperm cannot meet?', 'How does a surrogate mother help?'. They mark earlier questions answered by today's evidence (e.g. How does a mammal baby get nutrients?) and place unanswered ones in a 'Not	Say: 'Look at our board. Which questions can we now answer with evidence, and which new ones has today created?' Give WAIT TIME of 10-15 seconds before groups post. Cold-call 4 groups to read one question and justify its place. Highlight the question: 'What happens when natural fertilisation or pregnancy cannot occur?' and circle it as the bridge to Lesson 6. For the debate, set rules: 1 minute	Driving Question Board review: Making learners sort questions by 'answered', 'partly answered' and 'new' shows how evidence drives inquiry and gives purpose to the next lesson. Structured debate: Arguing from evidence requires learners to apply their understanding of pregnancy and care to a real social decision and to weigh different viewpoints respectfully.	Debate argument rubric (4 points each): claim stated clearly, biological evidence used, respectful rebuttal, and teamwork. Observable indicators: at least one correct biology fact per speaker (e.g. antenatal care reduces risks), no interruptions, and use of phrases such as 'the evidence shows'. DQB check: new questions are specific and testable, and at least one group

animals fertilisation implantation  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals fertilisation implantation	yet' column. They then take part in a short structured debate (6 minutes) on the statement: 'Decisions about pregnancy and reproductive health should be made jointly by both parents.' Two sides present evidence-based arguments with a rebuttal.	per speaker, use the biology evidence (e.g. antenatal visits, nutrition, birth planning), no personal stories or mocking, thank the opponent before rebuttal. Remind learners of the value of Peace: resolving differences amicably. Keep the discussion neutral and scientific, respecting diverse cultural backgrounds. Use the debate argument rubric while circulating.		poses a question about assisted reproduction.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: reproduction animals fertilisation implantation  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals fertilisation implantation	Groups revise the class model of 'How animals ensure successful fertilisation and development of offspring.' They add a new section for humans/viviparous mammals: internal fertilisation in the oviduct, implantation, placenta exchange, hormone control, birth and lactation. They compare it with the earlier frog (external fertilisation, many eggs, little care) and hen (shelled egg) sections using a comparison table: number of offspring, fertilisation type, nutrition source, parental care. Each group identifies the gap in the model: 'What happens when this natural process fails?' and writes it as a 'Model Gap' note. Individually they complete the exit ticket.	Say: 'Add today's evidence to our model. Every arrow must show a cause, and every claim must be backed by something we saw or heard.' Give WAIT TIME of 10-15 seconds before groups begin editing. Circulate and ask: 'Where does the baby's oxygen come from in your model? Show me the arrow.' Cold-call 3 groups to explain their revised model sections in 1 minute each. Prompt peer review: groups swap models and give one 'star' and one 'wish'. Point out that the model cannot yet explain what happens if the egg and sperm do not meet or if the uterus cannot support a pregnancy, and that this leads to Lesson 6 (IVF, surrogacy, sperm bank). Collect exit tickets and check them before the next lesson.	Concept Mapping/Hormonal Flowcharts and cumulative model revision: Merging today's mechanism into the class model connects structure, hormone and reproductive strategy across animals, and the 'Model Gap' note makes uncertainty explicit and motivates the next lesson. Peer review of models builds scientific critique.	Model checklist: includes placenta, umbilical cord, hormones (hCG, progesterone, oestrogen, oxytocin, prolactin), birth and lactation, and a comparison with frog and hen; at least 4 of 6 present. Peer review comments are specific. Exit ticket: (1) one thing the placenta does, (2) one hormone and its role, (3) one question about when reproduction fails. Look for at least 80% of learners giving a correct placenta function.

D. TEACHER REFLECTION

Which misconceptions persisted after the animations (e.g. mixing of maternal and fetal blood, IVF babies grown wholly in test tubes)? Did the guest speaker's input match the learning objectives and keep to respectful, scientific language? Were all learners, including those sharing devices, able to see and take part? Did the debate

stay neutral and respectful? What adjustments are needed for Lesson 6 on assisted reproduction (e.g. pre-teaching vocabulary such as embryo, surrogate, donor)?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed animations of fertilisation, implantation, placenta formation and birth, and heard a nurse/midwife explain antenatal care and stages of labour. We saw that the embryo implants in the uterine wall, the placenta forms, and the umbilical cord links the baby to the placenta.
What did I learn?	We learned that hormones (hCG, progesterone, oestrogen, oxytocin, prolactin) coordinate pregnancy, birth and lactation, and that the placenta exchanges oxygen, nutrients and wastes between mother and baby without mixing their blood.
How does this explain the phenomenon?	One well-nourished offspring (cow, human) is possible because the placenta and hormones provide sustained internal support, unlike frogs whose many eggs in water are unprotected. Our new question is: What happens when natural fertilisation or pregnancy cannot occur?

LESSON 6 (80 minutes (double lesson)): Helping Where Nature Needs a Hand: Debating IVF, Surrogacy and Sperm Banking

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use case studies to work out how reproductive biotechnologies (IVF, surrogacy, sperm banking) provide medical solutions to structural or hormonal infertility, and weigh scientific and ethical arguments through a structured debate.
Knowledge	By the end of the lesson the learner should be able to: f) appreciate the role of biotechnology in addressing human reproductive health; describe how IVF, surrogacy and sperm banking work and which cause of infertility each addresses (blocked fallopian tubes, low sperm count, hormonal imbalance, absent or unsafe uterus); explain that IVF only replaces fertilisation and early development outside the body, while the embryo is transferred to the uterus and develops there (not wholly in a test tube).
Skills	Developing and using models (flowchart of the IVF steps linked to hormones); engaging in argument from evidence (structured debate); obtaining, evaluating and communicating information from case studies; Think-Pair-Share discussion.
Attitudes	Core competencies: Critical thinking and Problem Solving, giving objective ideas to solve reproductive health problems while debating the role of biotechnology in human reproduction. Value: Peace, resolving differences amicably while debating. PCI: Human Sexuality, understanding common myths and misconceptions. Respect for diverse cultural and religious views and empathy for couples facing infertility.
Key Inquiry Question	How do animals reproduce? (Link: How can science assist when natural reproduction fails, and what do we need to consider before using it?)
Purpose in Storyline	Earlier lessons showed how frogs, fish, hens and cows achieve fertilisation and development, and how the human reproductive system and hormones work. This lesson returns to the final part of the driving question: how science steps in when human reproduction fails. It corrects the misconception that IVF babies grow in a test tube, and raises the next question: How do joint decisions by both parents affect family health and overall reproductive outcomes?
Safety Notes	No dissection in this lesson. Keep the debate neutral, scientific and respectful. Use fictional case studies so no learner or family is identified. Set ground rules: attack ideas, not people; no personal disclosure required; learners may pass. Be aware that some learners may have personal experience of infertility or adoption. Offer a private conversation or the school guidance and counselling teacher afterwards. Printed materials are used, and any video clips are pre-downloaded for offline use.

B. LESSON OVERVIEW

Learners begin by predicting what IVF involves and where the baby develops, using the frog, hen and cow comparison to anchor the idea of external versus internal fertilisation. In small groups they study case studies of fictional Kenyan couples (blocked fallopian tubes in Nakuru, low sperm count in Kisumu, hormonal imbalance in Eldoret, cancer treatment needing sperm banking in Nairobi, absent uterus considering surrogacy in Mombasa) and prepare for a structured debate. They then explain, using a hormonal flowchart, how each technology fixes a specific problem, add to the Driving Question Board, and revise the class model to include assisted reproduction as a human intervention that replicates steps that animals do naturally.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction animals ivf vitro	Learners recall the phenomenon: frogs release thousands of eggs into Lake Victoria wetlands, hens lay shelled eggs, cows give birth to one calf, and Kenyan news reports IVF babies. Individually they write answers to: (1) Where does fertilisation happen in IVF? (2) Where does the baby develop in the weeks after? (3) Which kinds of infertility could IVF solve and which could it not? Then Think-Pair-Share with a partner and record a prediction on a sticky note or in their notebooks.	Display a drawing of a frog, a hen, a cow and a couple. Say: "Many people believe IVF babies grow entirely in a test tube. Write down what you think happens, step by step." Give 2 minutes of silent writing, then WAIT TIME 10-15 seconds after each question. Pair learners for 2 minutes, then cold-call 4 pairs: "What did your partner predict, and why?" Record all ideas on the board without correcting. Say: "Hold on to your prediction. We will test it against case evidence." Remind the class of the ground rules for respectful discussion of sensitive topics.	Predict-Observe-Explain with Think-Pair-Share. Predicting surfaces the test-tube misconception and links to earlier ideas on external fertilisation (frogs) and internal fertilisation (cows), so learners later compare their predictions with evidence and see where their thinking changed.	Check prediction notes. Look for: whether learners place fertilisation outside the body in IVF, whether they identify the uterus as the site of development, and whether they name causes of infertility. Tally on the board how many learners think the baby develops wholly in a test tube, to compare with the exit ticket.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: reproduction animals ivf vitro	In groups of 4-5 (sharing printed copies), learners read one case study each: (A) A woman in Nakuru with blocked fallopian tubes: eggs cannot meet sperm. (B) A man in Kisumu with a very low sperm count. (C) A woman in Eldoret with irregular ovulation due to hormonal imbalance (low FSH and LH). (D) A young man in Nairobi about to begin cancer treatment that may damage his testes, considering a sperm bank. (E) A woman in Mombasa born without a functional uterus, considering surrogacy. Learners complete an evidence table: cause of infertility (structural/hormonal), technology suggested, steps involved, scientific benefits and limits, ethical questions (cost, consent, rights of the child, cultural and religious views, access in rural Kenya). Groups then hold a	Distribute case studies and the debate rubric. Say: "Your job is to use evidence from the case, not personal opinion alone." Circulate and ask: "Which part of the reproductive system is affected in this case?" "Which hormone or structure does this technology replace or bypass?" "What evidence supports your argument?" Allocate sides by lot, so learners argue positions they may not hold, and rotate speakers so everyone speaks. Enforce a timekeeper. WAIT TIME 10-15 seconds after each rebuttal question. Cold-call at least 6 learners across the debate: "Can you restate the opposing argument fairly before you respond?" Intervene calmly if language becomes disrespectful. Say: "We disagree with ideas, and we respect people." Clarify	Structured debate (argument from evidence). Assigned roles and a rubric force learners to separate scientific claims (what the technology does) from ethical claims (whether and how it should be used), so they build deeper understanding by using case evidence to support or challenge positions and by listening to opposing views.	Use the debate argument rubric (claim, evidence from case, scientific accuracy, rebuttal, respectful language) scored 1-4. Observable indicators: learners link technology to the correct cause of infertility (e.g., IVF to blocked tubes, hormone therapy to low FSH/LH), cite at least one piece of case evidence, and restate opposing points accurately. Note-takers' sheets are checked for at least 2 scientific and 2 ethical arguments.

	structured debate on the motion: "Reproductive biotechnologies are a responsible solution to infertility in Kenya." Format: opening statement (1 min each side), two rebuttal rounds, a closing statement, and a note-taker recording the main scientific and ethical arguments. Optional pre-downloaded 3-minute animation of IVF if devices are available.	misconceptions: IVF fertilises outside, but the embryo is transferred to the uterus; surrogacy involves a woman carrying the embryo; sperm banks freeze and store sperm.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: reproduction animals ivf vitro  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: reproduction animals ivf vitro	Learners return to their predictions and, in pairs, build a concept map or hormonal flowchart of IVF: hormone injections (FSH) stimulate several eggs to mature; eggs are collected; sperm is collected (or taken from a sperm bank); fertilisation in a dish; early embryo grows for a few days; embryo is transferred to the uterus; implantation; pregnancy continues in the uterus (placenta). They annotate which step replaces which natural step and compare with external fertilisation in frogs and internal fertilisation in cows. They then write a CER (Claim-Evidence-Reasoning) paragraph: "IVF babies do not grow wholly in a test tube because..." Finally learners match each case (A-E) to the best technology and explain why some cases need more than one option.	Say: "Look back at your prediction. What would you change, and what evidence made you change it?" Give WAIT TIME 10-15 seconds. Model the first two flowchart steps on the board, then let pairs complete the rest. Cold-call 5 pairs to explain one step: "Which natural step does this replace?" Prompt comparisons: "Frogs fertilise in water. How is IVF fertilisation similar, and how is it different?" and "Why does the embryo still need a uterus and placenta?" Address myths: infertility is not always the woman's problem; male factors account for a large share of cases; IVF does not guarantee pregnancy. Invite two learners to share their CER paragraphs.	Concept Mapping / Hormonal Flowcharts with Claim-Evidence-Reasoning. Sequencing the steps and linking each to a natural process and hormone makes the cause-and-effect chain visible, so learners connect biotechnology to the reproductive anatomy and hormones studied earlier and replace the test-tube misconception with an accurate model.	Check flowcharts for correct sequence (stimulation, egg collection, fertilisation in a dish, embryo transfer, implantation in the uterus), correct hormone labels, and a CER paragraph that states the uterus is the site of development. Use an exit ticket: (1) Name one cause of infertility and the technology that addresses it. (2) Give one scientific and one ethical consideration. (3) Where does the baby develop in IVF?
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms:	Learners revisit the class DQB. They add new evidence cards to the questions about how science assists when reproduction fails, move answered questions to the "answered" section, and write new questions arising from the debate, for example: Who makes decisions about treatment, and how do both parents affect the health of the child? What lifestyle factors affect	Say: "What did today's cases help us answer on our board? What new questions did the debate raise?" WAIT TIME 10-15 seconds. Cold-call 4 learners to place a card and justify it. Highlight the question: "How do joint decisions by both parents affect family health and overall reproductive outcomes?" and tell learners it will guide the next	Driving Question Board review. Tracking answered and new questions makes progress visible, links the lesson to the overall driving question, and gives learners ownership of the direction of inquiry.	Observe whether new questions are specific and build on evidence (e.g., ask about joint parental health or male fertility rather than a vague "is IVF good?"). Record the number of groups posting a question tied to the next lesson focus.

reproduction animals ivf vitro  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals ivf vitro	sperm and egg quality? How accessible is IVF in counties outside Nairobi? Each group posts at least one new question on the DQB.	lesson. Sort questions into the categories: biology, ethics, access and cost, and family health.		
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: reproduction animals ivf vitro  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: reproduction animals ivf vitro	Learners revise the class model of animal reproduction by adding a new component: "human intervention when natural reproduction fails." In groups they extend the earlier comparison of frogs, hens and cows and humans with a column for IVF, surrogacy and sperm banking, showing where fertilisation occurs, where the embryo develops, and which problems each addresses. They add arrows to show hormones (FSH, LH, oestrogen, progesterone) and draw a feedback loop showing how hormone imbalance can lead to infertility. They complete the summary table and a one-sentence statement of what changed in their thinking.	Say: "Add one new idea to our model and justify it with evidence from today." Circulate and ask: "What does your model now explain that it could not before?" "What does it still not explain?" WAIT TIME 10-15 seconds. Cold-call 4 groups to present their revised model for 1 minute each, and invite peer questions. Close by affirming that science offers options, but decisions involve health, ethics, culture and consent, and prompt learners to think about how both partners share responsibility for reproductive health.	Cumulative model revision with gallery review. Learners integrate the new evidence into the existing model and compare natural strategies across animals with assisted strategies in humans, so the model explains both how fertilisation and development are ensured and how science can help.	Check that models include: location of fertilisation and development for IVF, role of the uterus, the link between the type of infertility and the technology, and at least one hormone. Use peer review with the criteria: accurate, complete, evidence-based, clearly labelled. Collect exit tickets to identify learners who still hold the test-tube misconception.

D. TEACHER REFLECTION

After the lesson reflect on: Did learners move from the idea that IVF babies grow in a test tube to an evidence-based explanation? Did the debate remain respectful and scientifically grounded, and were any learners uncomfortable? Did every learner speak, or did a few dominate? Which cases (A-E) generated confusion about structural versus hormonal causes? What adjustments are needed for the next lesson, in which learners explore joint parental decisions and family health, for example by inviting the school nurse or a health worker as a resource person?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We read case studies of couples with blocked fallopian tubes, low sperm count, hormonal imbalance, a need to store sperm before cancer treatment, and an absent uterus, and we heard scientific and ethical arguments in a structured debate.
What did I learn?	IVF, surrogacy and sperm banking are biotechnologies that address specific structural or hormonal causes of infertility. In IVF fertilisation happens in a dish, but the embryo is transferred to the uterus and develops there. These options raise ethical questions about cost, consent, access and culture.
How does this explain the phenomenon?	Just as frogs, hens and cows have different strategies to ensure fertilisation and development, humans can use biotechnology to replace or bypass the step that fails (e.g., fertilisation when tubes are blocked), but the embryo still needs the uterus and hormones to develop, so the baby does not grow wholly in a test tube.

LESSON 7 (2 lessons of 40 minutes each (80 minutes)): Shared Decisions, Hormonal Health and Biotechnology: Our Final Explanation of How Animals Reproduce

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate the unit by explaining how animals ensure successful fertilisation and development of offspring, and how shared decision-making, hormonal health awareness and informed use of biotechnology support human reproductive health, then to compare Lesson 1 models with final models.
Knowledge	By the end of the lesson the learner should be able to: (a) describe external fertilisation in animals (frogs, fish) and types of internal fertilisation (oviparous, viviparous, ovoviviparous); (b) relate the structure of male and female reproductive systems to their functions; (c) describe fertilisation, pregnancy and birth in humans; (d) describe the role of hormones in human reproduction, including effects of undersecretion and oversecretion; (e) appreciate the role of biotechnology (surrogacy, sperm bank, in-vitro fertilisation) in addressing human reproductive health.
Skills	Constructing evidence-based explanations (NGSS SEP: Constructing Explanations); Developing and Using Models (comparing Lesson 1 and final models); Engaging in Argument from Evidence (structured debate); Analysing and Interpreting Data (reviewing dissection drawings and classification charts); Obtaining, Evaluating and Communicating Information (guest speaker Q&A). Core competencies: Critical Thinking and Problem Solving (giving objective ideas to solve reproductive health problems while debating the role of biotechnology in human reproduction); Learning to Learn (seeking information from the resource person).
Attitudes	Values: Peace (resolving differences amicably while debating biotechnology in human reproduction); Social Justice (sharing responsibilities fairly in group tasks). PCIs: Menstrual Health Disorders Management (sensitised on human menstrual health while illustrating the role of hormones in the human sexual cycle); Human Sexuality (understanding common myths and misconceptions around human sexuality). Appreciation of the role of biotechnology in human reproductive health, respect for animal welfare and for diverse cultural views.
Key Inquiry Question	How do animals reproduce?
Purpose in Storyline	This is the unit capstone. Learners have gathered evidence across Lessons 1 to 6 on external and internal fertilisation, reproductive structures, human fertilisation, pregnancy, birth, hormones and biotechnology. In Lesson 7 they argue from this evidence about joint reproductive choices, close out the Driving Question Board, and write a Final Explanation of the anchoring phenomenon: frogs with thousands of eggs, hens with shelled eggs, cows with one calf, and IVF/surrogacy in Kenya. They compare their Lesson 1 model with a final model and correct the misconception that IVF babies grow wholly in a test tube. No new question is raised; this is the unit reflection.
Safety Notes	Dissection (Predict-Observe-Explain review of the small mammal specimen, if still available, or of annotated photographs and drawings): use only specimens obtained under school and animal welfare guidelines; wear gloves, lab coats and eye protection; use sharp tools only under teacher supervision; keep specimens on trays; wash hands with soap after; dispose of tissue as directed by the school (burial or sealed disposal), never in open bins. Where a specimen is not available, use pre-downloaded photographs or drawings. Use no personal or intimate questions in the debate; participants argue positions assigned by the teacher, not their own family circumstances. Set ground rules: respect for cultural and religious views, no mocking, correct scientific terms, anonymous question box for sensitive questions. Pre-download all clips because internet access varies; learners share textbooks and devices in groups of 4 to 5.

B. LESSON OVERVIEW

Learners begin by predicting how well they can now explain the opening phenomenon (frogs near Kisumu laying thousands of eggs, hens laying shelled eggs, cows bearing one calf, and IVF babies in Kenyan news). They revisit the dissected small mammal (or annotated photographs) in a Predict-Observe-Explain review to link structure to function, then take part in a structured debate on joint reproductive health choices and the responsible use of IVF, surrogacy and sperm banks, using hormone flowcharts as evidence. A guest speaker (midwife, nurse or clinical officer from a local health centre, or a recorded interview) answers learner questions. Learners then review and close the Driving Question Board, compare their Lesson 1 model to a final synthesis model, and write a Final Explanation that answers the driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction animals joint reproductive	Learners retrieve their Lesson 1 initial model from their notebooks (the drawing of frog, hen, cow and IVF). In groups of 4, they take 2 minutes to write a prediction on a sticky note: 'If a Kisumu couple asked us how IVF works and why frogs lay thousands of eggs but a cow has one calf, what 3 key ideas would we tell them?' They also predict what the dissected mammal's reproductive organs will show about how internal fertilisation and development work (for example, where fertilisation and development occur). Then in a Think-Pair-Share they rate their own confidence (1 to 4) on four statements: external vs internal fertilisation, role of hormones, IVF and the 'test tube baby' idea, and the value of joint decisions by both parents.	Display the three images of the phenomenon (frog spawn in a Lake Victoria wetland, hen with eggs, cow with calf) and a Kenyan news headline about IVF. Say: 'In Lesson 1 you wrote what you thought. Today we close the unit. Take out your first model. Silently look at it for 1 minute.' WAIT TIME 10-15 seconds after asking: 'What would you change in it now, and why?' Give 2 minutes for writing, then 2 minutes for pair sharing. Cold-call 4 learners from different groups: 'Mwajuma, what is one idea you would keep? Otieno, what is one you would remove?' Record the common ideas on the board under 'Then' and leave space for 'Now'. Say: 'Remember, a prediction is not a test, it is our thinking made visible.' Note the misconception 'IVF babies grow wholly in a test tube' by tallying a show of hands without commenting yet.	Predict-Observe-Explain with Think-Pair-Share: writing predictions and rating confidence makes prior ideas visible and commits learners to a claim, so they can later see how and why their thinking changed. Retrieving the Lesson 1 model sets up the final comparison.	Observable indicators: (1) Predictions mention at least two ideas, for example 'external fertilisation needs water and many eggs because many are lost' or 'hormones control the cycle'. (2) Confidence ratings are recorded for all four statements. (3) Count how many learners still say IVF babies grow in a test tube; record the number (expected to be small, to be checked at exit). Teacher notes which groups cannot name any hormone so as to target them in the debate preparation.
Observe Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)	Station 1 (15 min, groups rotate in a Predict-Observe-Explain review): learners observe the small mammal specimen (teacher-demonstrated or photographs/annotated drawings from Lesson 4) and compare their annotated dissection drawings with	Before the specimen review, state the safety rules: 'Gloves on, tools stay on the tray, and we handle the specimen with respect. If you feel unwell, step back and tell me.' Circulate with the question: 'Which structure do you see here, and what does your drawing say it	Predict-Observe-Explain (specimen and animation) connects structures to functions with real evidence. Structured debate (Argument from Evidence) makes learners use claims, evidence and reasoning, hear counter-arguments and resolve	Observable indicators: (1) Annotated dissection drawings have correct labels for at least 5 structures with a function note each. (2) Peer review checklist on classification charts: animals correctly grouped (frog and tilapia as external; hen as oviparous; cow

Search ARES for similar videos READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: reproduction animals joint reproductive	the labelled chart of the male and female reproductive systems. They identify structures (ovaries, oviducts, uterus, testes, sperm duct) and match each to a function. Station 2 (10 min): learners use a pre-downloaded animation clip on fertilisation, implantation, placenta and birth and a classification chart to confirm where fertilisation and development occur in frogs and fish (external), hens (oviparous, internal fertilisation), cows and humans (viviparous), and some sharks or snakes (ovoviviparous). Peer review: partner groups swap classification charts and use a checklist to mark accuracy. Debate preparation (15 min): groups are assigned positions on the motion 'Reproductive health choices should be made jointly by both parents, and medical technology such as IVF, surrogacy and sperm banks should be used with informed consent.' Positions: Group A (for joint decisions), Group B (cautions and limits), Group C (role of medical technology), Group D (hormonal health awareness, including undersecretion and oversecretion). Then the structured debate (20 min). Finally, a guest speaker (midwife or nurse; or a recorded interview if no visitor) answers 5 prepared learner questions on pregnancy, childbirth, postnatal health and fertility services (15 min).	does?' WAIT TIME 10-15 seconds after each question before taking answers. Cold-call 3 groups to explain one structure-function link (for example, 'Why is the uterus muscular and thick-walled?'). For the animation: pause at the placenta and ask, 'What passes across here, and what does that tell us about the idea that IVF babies grow in a test tube?' Cold-call 2 learners. For the debate, set the rules: 'Speak for 1 minute, use evidence from our lessons, and begin with: I claim..., my evidence is..., this means...' Appoint a timekeeper and a neutral scribe per group. Intervene with: 'Let us focus on the biology and keep the tone respectful.' Supply hormone flowcharts (FSH, LH, oestrogen, progesterone, oxytocin, prolactin, testosterone) as evidence; prompt: 'What would happen to the menstrual cycle if oestrogen were undersecreted? What if there were oversecretion?' Check the anonymous question box before the guest speaker and remove inappropriate items. For the speaker: 'Raise your hand; ask one clear question; we listen first.' Cold-call 5 learners to ask the prepared questions and use follow-up oral probes ('Can you say more about what you heard?').	differences amicably (Peace). Concept Mapping/Hormonal Flowcharts help learners see feedback and cause-and-effect chains, such as how hormonal imbalance may affect ovulation or sperm production and why a couple may seek IVF.	and human as viviparous; some sharks/snakes as ovoviviparous) with 80% accuracy. (3) Debate argument rubric (4 levels) scoring claim, scientific evidence, reasoning, and respect; learners at level 3 or 4 cite at least two pieces of evidence from earlier lessons. (4) Oral probes during guest speaker Q&A: learners ask relevant, clear questions and can paraphrase an answer. (5) Teacher records which learners still confuse hormone roles.
Explain Phase VIDEO: Plant reproductive success Source: Khan	Individually (5 min), then in pairs (5 min), learners draft the Final Explanation answering the driving question using a Claim-Evidence-Reasoning frame with four required parts: (1) why frogs and	Say: 'You are now the experts. Your task is to explain the whole phenomenon in your own words, using evidence.' Display the frame: 'Claim: ... Evidence from our lessons: ... Reasoning: ...'. Pose:	Claim-Evidence-Reasoning within Think-Pair-Share: writing and then talking to peers forces learners to link observations to scientific principles (fertilisation site, number of offspring, parental care,	Observable indicators: a successful Final Explanation (a) names external and internal fertilisation and links each to egg number and care; (b) uses at least two hormone examples correctly;

Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: reproduction animals joint reproductive	fish use external fertilisation and produce thousands of eggs (water needed, eggs and sperm released into water, no parental care, high losses to predators), whereas cows and humans use internal fertilisation with few offspring and parental care; (2) how structure fits function in male and female reproductive systems and how hormones coordinate cycles, pregnancy, birth and lactation; (3) how IVF, surrogacy and sperm banks help when reproduction fails: in IVF fertilisation occurs outside the body but the embryo develops in the uterus, which corrects the misconception; (4) why joint decisions by both parents and hormonal health awareness matter. Then pairs share with another pair (Think-Pair-Share) and refine.	'Why would a frog near Kisumu need thousands of eggs but a cow only one calf?' WAIT TIME 10-15 seconds before pairs talk. Walk around and ask: 'Which piece of evidence supports that? Which animal does your idea not explain?' Cold-call 4 pairs to read one sentence of their explanation; ask the class to give thumbs up/sideways/down for 'evidence-based?' Press on the IVF idea with: 'Where does the embryo grow after IVF? What evidence do we have?' Address myths about hormones, menstruation and sexuality with neutral, scientific language: 'Menstruation is a normal biological process controlled by hormones.' Provide sentence starters for learners who need support. Offer an extension: 'How would the explanation change for a tilapia farmer in Kisumu who wants more fish?'	hormone control, medical intervention). Hearing alternative wording strengthens conceptual understanding and exposes gaps.	(c) states that after IVF the embryo develops in the mother's or surrogate's uterus; (d) states at least one reason joint reproductive decisions matter. Use a 3-point checklist (all four present = 3; two or three = 2; fewer = 1). Teacher records learners scoring 1 for follow-up. Look for correct vocabulary: external fertilisation, oviparous, viviparous, ovoviviparous, placenta, embryo.
Driving Question Board (DQB) Creation VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: reproduction	Learners review the class Driving Question Board built across the unit. In groups of 4 they take question cards (for example, 'Why do frogs lay so many eggs?', 'What is the difference between oviparous and viviparous?', 'Is IVF baby grown in a test tube?', 'What do hormones do during pregnancy?', 'Why is it important for both parents to make reproductive choices together?'). Each group sorts cards into 'Answered with evidence', 'Partly answered' and 'Still open', and writes the answer and the lesson where the evidence was found on a sticky note. Each group also adds one new 'future question' (for example, about fertility in livestock, or health services in their county) to a 'Beyond the unit' column.	Say: 'Our board holds every question we asked. Let us see how many we can now answer with evidence.' Give each group 5 cards and 8 minutes. Ask: 'Which lesson gave you the evidence? What did you see, do or read?' WAIT TIME 10-15 seconds before taking answers. Cold-call 5 groups to read a card and the evidence-based answer; invite others to challenge with 'I would add...' Ensure fairness with 'Every member must speak once' (Social Justice). Place 'still open' cards in the 'Beyond the unit' column and say: 'Real scientists in Kenya, such as researchers at KALRO or ILRI in Nairobi, also have open questions; this is how science continues.' Check for any unanswered sensitive question left	Driving Question Board review: learners see how their questions led the learning, link each answer to evidence and recognise that science is an ongoing process. Sorting and justifying builds metacognition and ownership of learning.	Observable indicators: at least 80% of cards are placed in the correct category with a lesson reference; sticky-note answers use scientific terms accurately; groups can explain one question that moved from 'open' to 'answered'; new questions are testable or researchable. Exit ticket (on a half sheet): 'Write one thing you now understand that you did not at the start, and one question you still have.' Teacher sorts exit tickets into 'secure' and 'needs revisiting'.

animals joint reproductive		in the anonymous box and answer it briefly and factually, or refer it to the health teacher or guest speaker.		
Model Building Phase  VIDEO: Plant reproductive success Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: reproduction animals joint reproductive	Learners take their Lesson 1 model and build a final synthesis model on a large sheet, as a concept map or labelled diagram, with three linked sections: (1) a comparison of frog (external fertilisation, thousands of eggs), hen (internal fertilisation, shelled eggs) and cow/human (internal fertilisation, development in uterus, placenta, one or few offspring); (2) a hormonal flowchart for the human cycle, pregnancy, birth and lactation showing the effects of undersecretion and oversecretion; (3) a section on IVF, surrogacy and sperm banks showing where fertilisation and development take place, with the heading 'Shared decisions' showing informed consent and the roles of both parents. They use a different colour to mark what changed from the Lesson 1 model. In a gallery walk, groups give two 'stars' (strengths) and one 'step' (suggestion) on sticky notes using the reflection rubric. Finally, learners complete the individual summary table (observed, learned, explained) and a short written reflection.	Say: 'Place your first model beside your new one. A model is a tool for explaining; it must show what is happening and why.' Give success criteria: 'Your model must (1) show where fertilisation happens in each animal, (2) show at least three hormones and their functions, (3) show how IVF fits, (4) show what changed from Lesson 1.' Circulate with: 'What does this arrow mean? What evidence led you to add it?' WAIT TIME 10-15 seconds. Support with the prompt, 'What would a Kisumu farmer or a Nairobi family need to understand from this model?' Run a gallery walk for 8 minutes. Cold-call 4 groups to name one major change in their model since Lesson 1 (for example, 'We removed the idea that IVF babies grow wholly in a test tube'). Close with: 'Science also needs people who make good decisions, so what role do you play in your own reproductive health and in respecting others' choices?' Collect the summary tables and Final Explanations for review.	Concept Mapping/Hormonal Flowcharts and model revision: placing the initial and final models side by side makes conceptual change visible. Peer review with the rubric encourages evaluative thinking. Linking animal strategies, human hormones and technology in one model integrates the whole unit and answers the driving question.	Observable indicators using the reflection rubric: (a) model shows external vs internal fertilisation correctly with reasons; (b) hormonal flowchart has at least three hormones with correct roles and an example of undersecretion/oversecretion; (c) IVF shown as fertilisation outside the body with the embryo developing in the uterus; (d) a clear 'what changed' annotation; (e) shared decision-making is explained. Peer 'stars and steps' comments are specific. Exit ticket: learners state in one sentence how science can assist when human reproduction fails. Teacher records learners needing support.

D. TEACHER REFLECTION

After the lesson, consider: (1) How many learners moved from 'IVF babies grow wholly in a test tube' to the correct idea that the embryo develops in the uterus? Check the exit tickets and Final Explanations. (2) Did the debate stay scientific and respectful of diverse cultural views? Which ground rules worked and which need strengthening? (3) Could learners link the number of eggs to the site of fertilisation and parental care? (4) Which hormone roles remain unclear, and how will you revisit them? (5) Were safety and animal welfare guidelines followed during any specimen work, and were pre-downloaded clips accessible to all groups despite limited internet? (6) Did every group member contribute fairly (Social Justice)? (7) Which unanswered DQB questions might inform later units or a school health club visit?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What we observed: external fertilisation in frogs and fish (many eggs released into water), internal fertilisation in hens, cows and humans, the structures of the male and female reproductive systems in the specimen or drawings, the animation of fertilisation, pregnancy and birth, and arguments in the class debate and from the guest speaker about joint reproductive choices and biotechnology.
What did I learn?	What we learned: the types of fertilisation and development (external, oviparous, viviparous, ovoviviparous); how structure relates to function in reproductive systems; how hormones control the cycle, pregnancy, birth and lactation, and what undersecretion and oversecretion can cause; how IVF, surrogacy and sperm banks assist when reproduction fails; and why shared decisions by both parents matter.
How does this explain the phenomenon?	What we can now explain: why frogs near Kisumu lay thousands of eggs while a cow bears one calf (site of fertilisation, losses, parental care); why IVF babies do not grow wholly in a test tube (fertilisation occurs outside, but the embryo develops in the uterus); and how animals ensure successful fertilisation and development, and how science and shared, informed decisions support human reproductive health.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.